MATH2121: Linear Algebra

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1 Linear Systems and Matrices

1.1 Notation

- $\mathbb{R}$: real numbers.
- $\mathbb{Q}$: rational numbers p/q.
- $\mathbb{Z}$: integers $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
- $\mathbb{N}$: nonnegative integers $\{0, 1, 2, \ldots\}$.

1.2 Systems of Linear Equations

Definition. A *linear equation* in variables $x_1, x_2, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, \ldots, a_n, b \in \mathbb{R}$. Equivalently, $\sum_{i=1}^{n} a_ix_i = b$.

Example. The following are linear equations in x_1, x_2, x_3:

$$3x_1 = 2x_2, \quad 3x_1 + 4x_2 - \sqrt{2}x_3 = 7, \quad 0 = 0, \quad 0 = 1.$$

The following are **not** linear equations:

$$3x_1^2 + 4x_2 = 7, \quad x_1x_2 = x_3, \quad \sqrt{x_1} = x_2.$$

Definition. A *system of linear equations* (or *linear system*) is a list of linear equations in the same set of variables. A *solution* is a list $(s_1, \ldots, s_n)$ such that substituting $x_i = s_i$ makes all equations true. A solution is *nonzero* if at least one $s_i \neq 0$.

Definition. Two linear systems are *equivalent* if they have the same set of solutions.

Number of solutions

Theorem. A linear system (in any number of variables) has either 0, 1, or infinitely many solutions.

Proof (algebraic). Suppose the linear system has two distinct solutions $\vec{s} = (s_1, \ldots, s_n)$ and $\vec{r} = (r_1, \ldots, r_n)$, so $\vec{s} \neq \vec{r}$. Consider the associated *homogeneous system*, obtained by replacing each b_i with 0. For each equation $\sum_{j=1}^{n} a_{ij}x_j = b_i$, since $\vec{s}$ and $\vec{r}$ are both solutions we have $\sum_{j=1}^{n} a_{ij}s_j = b_i$ and $\sum_{j=1}^{n} a_{ij}r_j = b_i$. Subtracting gives $\sum_{j=1}^{n} a_{ij}(s_j - r_j) = 0$, so $\vec{s} - \vec{r} = (s_1 - r_1, \ldots, s_n - r_n)$ is a nonzero solution of the homogeneous system. Let $\vec{t} = \vec{s} - \vec{r} \neq \vec{0}$. For any scalar $c \in \mathbb{R}$, the vector $\vec{r} + c\vec{t}$ satisfies $\sum_{j=1}^{n} a_{ij}(r_j + ct_j) = b_i + c \cdot 0 = b_i$, so $\vec{r} + c\vec{t}$ is a solution. As c ranges over $\mathbb{R}$, the vectors $\vec{r}, \vec{r} + \vec{t}, \vec{r} + 2\vec{t}, \vec{r} + 3\vec{t}, \ldots$ are all distinct (since $\vec{t} \neq \vec{0}$), giving infinitely many solutions. Hence a system with at least two solutions has infinitely many. Therefore the only possibilities are 0, 1, or infinitely many solutions. ■

Proof (geometric). Each equation $\sum_{j=1}^{n} a_{ij}x_j = b_i$ defines a hyperplane in $\mathbb{R}^n$. The solution set is the intersection of all these hyperplanes. If the intersection is empty, there

are 0 solutions. If the intersection is a single point, there is 1 solution. If the intersection contains two distinct points $\vec{s}$ and $\vec{r}$, then the entire line $\{\vec{r} + c(\vec{s} - \vec{r}) : c \in \mathbb{R}\}$ lies in the intersection (each hyperplane, containing two points of the line, contains the whole line), giving infinitely many solutions. ■

Definition. A linear system is *consistent* if it has at least one solution, and *inconsistent* if it has zero solutions.

1.3 Matrices

Definition. A *matrix* is a rectangular array of numbers. A matrix is $m \times n$ (or m-by-n) if it has m rows and n columns. We write A_{ij} for the entry in the ith row and jth column.

Definition. Given a linear system

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

the *coefficient matrix* is $\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$ and the *augmented matrix* is

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{pmatrix}.$$

The coefficient matrix is $m \times n$; the augmented matrix is $m \times (n+1)$.

1.4 Elementary Row Operations

Definition. The *elementary row operations* on a matrix are:

1. **Replacement:** Replace a row by the sum of itself and a multiple of another row.
2. **Scaling:** Multiply a row by a nonzero number.
3. **Interchange:** Swap two rows.

Definition. Two matrices are *row equivalent* if one can be transformed to the other by a sequence of zero or more row operations.

Each row operation is reversible

- Scaling by $c \leftrightarrow$ scaling by $1/c$.
- Swapping rows i and $j \leftrightarrow$ swapping rows i and j.
- Replacing row i by (row i + $c\cdot$ row j) $\leftrightarrow$ replacing row i by (row i − $c\cdot$ row j).

Row equivalence preserves solutions

Theorem. If the augmented matrices of two linear systems are row equivalent, then the systems are equivalent (i.e., have the same solutions).

Proof. It suffices to show that a single elementary row operation on the augmented matrix produces a new system with the same solution set. There are three cases. Let the original augmented matrix correspond to the system $\sum_{j=1}^{n} a_{ij}x_j = b_i$ for $i = 1, \ldots, m$.

1. **Replacement:** Replace row i by (row i + $c\cdot$ row k) with $k \neq i$ and $c \in \mathbb{R}$. The ith

equation becomes $\sum_{j=1}^{n}(a_{ij} + c\,a_{kj})x_j = b_i + c\,b_k$. If $\vec{s}$ solves the original system, then $\sum a_{ij}s_j = b_i$ and $\sum a_{kj}s_j = b_k$, so $\sum(a_{ij} + c\,a_{kj})s_j = b_i + c\,b_k$, meaning $\vec{s}$ solves the new system. Conversely, if $\vec{s}$ solves the new system, then $\sum(a_{ij} + c\,a_{kj})s_j = b_i + c\,b_k$, and using $\sum a_{kj}s_j = b_k$ (the kth equation is unchanged) we recover $\sum a_{ij}s_j = b_i$. So the solution sets are equal.

2. **Scaling:** Multiply row i by $c \neq 0$. The ith equation becomes $\sum_{j=1}^{n}(c\,a_{ij})x_j = c\,b_i$, i.e., $c\sum a_{ij}x_j = cb_i$. Since $c \neq 0$, dividing by c recovers the original equation, so the solution sets are equal.
3. **Interchange:** Swap rows i and k. This merely reorders the equations, which does not change the solution set (a list $(s_1, \ldots, s_n)$ satisfies all equations regardless of their order).

Since each elementary row operation preserves the solution set, any sequence of them (i.e., row equivalence) preserves the solution set. ■

1.5 Echelon Form and RREF

Definition. A row in a matrix is *nonzero* if not every entry is zero. The *leading entry* in a nonzero row is the first nonzero entry (from left to right).

Definition. A matrix with m rows and n columns is in *echelon form* if:

(E1) When a row is nonzero, every row above it is also nonzero.
(E2) The leading entry in a nonzero row is strictly to the right of the leading entry of any earlier row.

These imply:

(E3) If a row is nonzero, then every entry below its leading entry in the same column is zero.

Definition. A matrix in echelon form is *reduced* if:

(R1) Each nonzero row has leading entry 1.
(R2) The leading 1 in each nonzero row is the only nonzero number in its column.

Such a matrix is said to be in *reduced echelon form.*

Uniqueness of RREF

Theorem. Each matrix A is row equivalent to exactly one matrix RREF(A) in reduced echelon form, called the *reduced echelon form* of A.

Proof. Existence follows from the row reduction algorithm, which produces a reduced echelon form. We prove uniqueness. Suppose A is row equivalent to two reduced echelon matrices R and S. Since R and S are both row equivalent to A, they are row equivalent to each other. We show $R = S$ by induction on the number of columns. The key idea: the pivot columns of any echelon form of A are in the same positions (this can be shown by induction on the number of pivot columns, since the leftmost nonzero column must contain a pivot in both R and S, and reducing the submatrix below that pivot column gives the same structure recursively). So R and S have pivots in the same positions. In a reduced echelon form, each pivot column is a standard basis vector $\vec{e}_i$ (with 1 in the pivot row and 0 elsewhere). Since R and S have the same pivot positions, each pivot column of R equals the corresponding pivot column of S. For a non-pivot column, its entries are determined: the entry in a pivot row equals the coefficient needed to express that column as a linear combination of the pivot columns to its left, which is uniquely determined by

the column itself and the pivot columns. Since both R and S are obtained from A by row operations (which preserve linear relations among columns), the non-pivot columns of R and S coincide. Therefore $R = S$. ■

1.6 Solving a Linear System from RREF

Definition. A *pivot position* in a matrix A is the location containing a leading 1 in RREF(A). A *pivot column* is a column containing a pivot position. An entry in a pivot position is called a *pivot*.

Definition. For a linear system in variables $x_1, \ldots, x_n$ with augmented matrix A, the variable x_i is *basic* if i is a pivot column of A, and *free* otherwise.

Solutions from RREF

Theorem. Consider a linear system whose augmented matrix is A.

- The system has **0 solutions** if the last column of A contains a pivot (i.e., RREF(A) has a row $(0 \quad \cdots \quad 0 \quad 1)$, giving the false equation $0 = 1$).
- The system has **1 solution** if there are no free variables and the last column is not a pivot.
- The system has **infinitely many solutions** otherwise (i.e., the last column is not a pivot but there is at least one free variable).

Proof. Row-reduce A to RREF(A) $= R$. By the previous theorem, R represents an equivalent system with the same solution set.

1. If the last column of R is a pivot column, then some row of R is $(0 \quad \cdots \quad 0 \quad 1)$, corresponding to the equation $0 = 1$, which has no solution. So the system is inconsistent: 0 solutions.
2. If the last column is not a pivot and there are no free variables, then every variable $x_1, \ldots, x_n$ is basic (each corresponds to a pivot column). Each basic variable is uniquely determined (e.g., $x_i = b_i$ for some determined value b_i), so there is exactly one solution.
3. If the last column is not a pivot but there is at least one free variable, then the system is consistent (no row $(0 \quad \cdots \quad 0 \quad 1)$) and each free variable can take any real value. For each assignment of the free variables, the basic variables are uniquely determined. Since there is at least one free variable ranging over $\mathbb{R}$, there are infinitely many solutions.

These three cases are exhaustive and mutually exclusive. ■

General solution procedure. After computing RREF(A):

1. Identify basic and free variables.
2. Write each basic variable in terms of the free variables.
3. Choose arbitrary values for free variables; solve for basic variables.

Corollary. If $\text{RREF}(A) = \begin{pmatrix} 1 & 0 & \cdots & 0 & b_1 \\ 0 & 1 & \cdots & 0 & b_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & b_m \end{pmatrix}$ then the unique solution is $(x_1, \ldots, x_m) = (b_1, \ldots, b_m)$.

Proof. The RREF shows each variable is a basic variable (every column $1, \ldots, m$ is a pivot column) and the last column is not a pivot. By the previous theorem, there is exactly one solution. Reading off the RREF, the ith equation is $x_i = b_i$ (the leading 1 in column i gives x_i plus zeros from other entries). So $x_i = b_i$ for all i, giving the unique solution $(x_1, \ldots, x_m) = (b_1, \ldots, b_m)$. ■

1.7 Row Reduction Algorithm

Algorithm (Row reduction to echelon form). Input: an $m \times n$ matrix A.

1. Begin with the leftmost nonzero column. This is a pivot column; the pivot position is the top position.
2. Select a nonzero entry in the current pivot column. If needed, swap its row to the top.
3. Use row operations to create zeros below the pivot position.
4. Cover the row containing the current pivot; apply steps 1–3 to the remaining $(m-1)\times n$ submatrix. Repeat until the entire matrix is in echelon form.
5. Starting from the rightmost pivot, rescale to leading entry 1, then create zeros *above* each pivot. Repeat going right to left.

Output: RREF(A).

Pivot positions are invariant

In any echelon form E of a matrix A, the locations of the leading entries are the same. Thus pivot positions can be found from *any* echelon form—one need not fully reduce to find pivots.

2 Vectors and Linear Combinations

2.1 Vectors

Definition. A *(column) vector* of size n is an $n \times 1$ matrix: $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$. We write $\mathbb{R}^n$ for the set of all vectors with exactly n rows. Each v_i is a real number.

Vector operations:

- *Addition* (same size only): $\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{pmatrix}$.
- *Scalar multiplication* ($c \in \mathbb{R}$): $c \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} cv_1 \\ \vdots \\ cv_n \end{pmatrix}$.
- *Subtraction*: $\vec{u} - \vec{v} = \vec{u} + (-1)\vec{v}$.

Definition. The *zero vector* in $\mathbb{R}^n$ is $\vec{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$. We always have $\vec{0} + \vec{v} = \vec{v} + \vec{0} = \vec{v}$.

Geometric intuition in $\mathbb{R}^2$

A vector $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is drawn as an arrow from the origin to (a_1, a_2). The sum $\vec{a} + \vec{b}$ is the diagonal of the parallelogram with sides $\vec{a}$ and $\vec{b}$.

2.2 Linear Combinations and Span

Definition. If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p \in \mathbb{R}^n$ and $c_1, \ldots, c_p \in \mathbb{R}$ (scalars), then

$$\vec{y} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p$$

is a *linear combination* of $\vec{v}_1, \ldots, \vec{v}_p$ with coefficients $c_1, \ldots, c_p$.

Definition. The *span* of vectors $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ is the set of all their linear combinations:

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} = \{c_1\vec{v}_1 + \cdots + c_p\vec{v}_p : c_1, \ldots, c_p \in \mathbb{R}\} \subseteq \mathbb{R}^n.$$

Span $\Leftrightarrow$ consistent system

Proposition. A vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$ (where $\vec{a}_1, \ldots, \vec{a}_n, \vec{b} \in \mathbb{R}^m$) has the same solutions as the linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$.

Proof. Write each vector $\vec{a}_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$. The vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$ means, in the ith coordinate, $\sum_{j=1}^{n} a_{ij}x_j = b_i$ for $i = 1, \ldots, m$. But these are exactly the m equations of the linear system with coefficient matrix (a_{ij}) and augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$. Therefore a list $(s_1, \ldots, s_n)$ satisfies the vector equation if and only if it satisfies each of the m scalar equations, i.e., it is a solution of the linear system. The two have identical solution sets. ■

Corollary. $\vec{b} \in \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ if and only if the matrix $\begin{pmatrix} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_p & \vec{y} \end{pmatrix}$ is the augmented matrix of a consistent linear system (i.e., the last column is not a pivot column).

Proof. By definition, $\vec{b} \in \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ if and only if there exist scalars $x_1, \ldots, x_p$ with $\vec{b} = x_1\vec{v}_1 + \cdots + x_p\vec{v}_p$, i.e., the vector equation has a solution. By the proposition, this is equivalent to the linear system with augmented matrix $\begin{pmatrix} \vec{v}_1 & \cdots & \vec{v}_p & \vec{b} \end{pmatrix}$ having a solution, i.e., being consistent. By the RREF criterion, a system is consistent if and only if the last column of the augmented matrix is not a pivot column (no row $\begin{pmatrix} 0 & \cdots & 0 & 1 \end{pmatrix}$). ■

Span of a single vector

$\operatorname{span}\{\vec{0}\} = \{\vec{0}\}$ (a single point). For $\vec{v} \neq \vec{0}$, $\operatorname{span}\{\vec{v}\} = \{c\vec{v} : c \in \mathbb{R}\}$ is the line through the origin in the direction of $\vec{v}$. This is infinite unless $\vec{v} = \vec{0}$.

Geometric picture of span.

- In $\mathbb{R}^2$: span is a point, a line through origin, or all of $\mathbb{R}^2$.
- In $\mathbb{R}^3$: span is a point, a line through origin, a plane through origin, or all of $\mathbb{R}^3$.

3 Matrix Equations and Matrix Operations

3.1 Matrix-Vector Product

Definition. If A is an $m \times n$ matrix with columns $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n \in \mathbb{R}^m$ and $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$, then the *matrix-vector product* $A\vec{v}$ is the vector in $\mathbb{R}^m$ given by:

$$A\vec{v} = \begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = v_1\vec{a}_1 + v_2\vec{a}_2 + \cdots + v_n\vec{a}_n \in \mathbb{R}^m.$$

Thus $A\vec{v}$ is the linear combination of the columns of A with coefficients from $\vec{v}$.

Dimensions and linearity

- $A\vec{v}$ is defined only when #columns of A = #rows of $\vec{v}$. If A is $m \times n$ and $\vec{v} \in \mathbb{R}^n$, then $A\vec{v} \in \mathbb{R}^m$.
- A transforms $\mathbb{R}^n \to \mathbb{R}^m$. This transformation is **linear**:
  1. $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$ for all $\vec{u}, \vec{v} \in \mathbb{R}^n$.
  2. $A(c\vec{v}) = c(A\vec{v})$ for all $\vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Entry-wise formula. The ith entry of $A\vec{v}$ is the dot product of row i of A with $\vec{v}$:

$$(A\vec{v})_i = a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n.$$

3.2 Matrix Equations

Definition. If A is $m \times n$, $\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ (variables), and $\vec{b} \in \mathbb{R}^m$, then $A\vec{x} = \vec{b}$ is a *matrix equation.*

Three equivalent views

Proposition. The matrix equation $A\vec{x} = \vec{b}$ has the same solutions as:
1. The vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$.
2. The linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$.

Proof. By definition of the matrix-vector product, $A\vec{x} = \begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$. So $A\vec{x} = \vec{b}$ is precisely the vector equation $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{b}$, and they have identical solutions. By the proposition in Section 2, the vector equation has the same solutions as the linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$. Therefore all three formulations share the same solution set. ■

Proposition. $A\vec{x} = \vec{b}$ has a solution $\Leftrightarrow$ $\vec{b}$ is a linear combination of the columns of A $\Leftrightarrow$ $\vec{b} \in \text{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$.

Proof. $A\vec{x} = \vec{b}$ has a solution $\Leftrightarrow$ there exist $x_1, \ldots, x_n$ with $A\vec{x} = \vec{b}$ $\Leftrightarrow$ there exist $x_1, \ldots, x_n$ with $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{b}$ (by the definition of matrix-vector product) $\Leftrightarrow$ $\vec{b}$ is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ $\Leftrightarrow$ $\vec{b} \in \text{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ (by the definition of span). ■

Four equivalent conditions

Theorem. Let A be an $m \times n$ matrix. The following are equivalent:

1. For each $\vec{b} \in \mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.
2. Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
3. The span of the columns of A is $\mathbb{R}^m$ ("the columns of A span $\mathbb{R}^m$").
4. A has a pivot position in every row.

Proof. We show (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d).

(a) $\Leftrightarrow$ (b): By the previous proposition, $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A. So (a) (for each $\vec{b}$, $A\vec{x} = \vec{b}$ has a solution) is equivalent to (b) (each $\vec{b}$ is a linear combination of the columns of A).

(b) $\Leftrightarrow$ (c): Saying each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A is exactly saying $\mathbb{R}^m \subseteq \text{span}\{\text{columns of } A\}$. Since the columns of A are in $\mathbb{R}^m$, the span is a subset of $\mathbb{R}^m$. Hence $\text{span}\{\text{columns}\} = \mathbb{R}^m$, which is (c).

(c) $\Leftrightarrow$ (d): Suppose (c): the columns of A span $\mathbb{R}^m$. Then for each $\vec{b} \in \mathbb{R}^m$, the system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$ is consistent, so the last column is never a pivot column. This means no row of $\text{RREF}(A)$ is all zeros (otherwise appending $\vec{b} = \vec{e}_i$ for the corresponding row would make the last column a pivot), so every row of $\text{RREF}(A)$ has a pivot: (d). Conversely, suppose (d): every row of A has a pivot. Then $\text{RREF}(A)$ has no zero rows, so for any $\vec{b}$, the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ cannot have a pivot in the last column (a pivot in the last column would require a zero row in the coefficient part, contradicting every row having a pivot). So the system is always consistent, giving (c). ■

3.3 Sums and Scalar Multiples

Definition. If $T, U : \mathbb{R}^n \to \mathbb{R}^m$ are linear with the same domain and codomain:

- *Sum:* $(T + U)(\vec{v}) = T(\vec{v}) + U(\vec{v})$.
- *Scalar multiple:* $(cT)(\vec{v}) = c\,T(\vec{v})$ for $c \in \mathbb{R}$.

Both $T + U$ and cT are linear transformations.

Definition. If $A = (a_{ij})$ and $B = (b_{ij})$ are $m \times n$ matrices:

- $A + B = (a_{ij} + b_{ij})$ (entry-wise sum, same size required).
- $cA = (c\,a_{ij})$ (scalar multiplication).

Proposition. The standard matrix of $T + U$ is $A + B$; the standard matrix of cT is cA.

Proof. Let A and B be the standard matrices of T and U respectively, so $T(\vec{v}) = A\vec{v}$ and $U(\vec{v}) = B\vec{v}$ for all $\vec{v}$. Then for every $\vec{v} \in \mathbb{R}^n$, $(T+U)(\vec{v}) = T(\vec{v}) + U(\vec{v}) = A\vec{v} + B\vec{v} = (A + B)\vec{v}$ (using entry-wise definition of matrix addition and linearity of matrix-vector product). So $A+B$ is a matrix representing $T+U$. By uniqueness of the standard matrix, $A + B$ is the standard matrix of $T + U$. Similarly, $(cT)(\vec{v}) = c\,T(\vec{v}) = c(A\vec{v}) = (cA)\vec{v}$ (using entry-wise scalar multiplication), so cA is the standard matrix of cT. ■

Properties of matrix addition. For matrices of the same size and $a, b \in \mathbb{R}$:

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.

3. $A + 0 = A$ (zero matrix).
4. $a(A + B) = aA + aB$.
5. $(a + b)A = aA + bA$.
6. $a(bA) = (ab)A$.

Proof: Let $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$ be $m \times n$ matrices, and let 0 denote the $m \times n$ zero matrix. We prove each property entry-wise.

1. *Commutativity* $A+B = B+A$: The (i, j) entry of $A+B$ is $a_{ij}+b_{ij}$, and the (i, j) entry of $B+A$ is $b_{ij}+a_{ij}$. Since addition of real numbers is commutative, $a_{ij}+b_{ij} = b_{ij}+a_{ij}$ for all i, j. Hence $A + B = B + A$.
2. *Associativity* $(A+B)+C = A+(B+C)$: The (i, j) entry of $(A+B)+C$ is $(a_{ij}+b_{ij})+c_{ij}$, and the (i, j) entry of $A+(B+C)$ is $a_{ij}+(b_{ij}+c_{ij})$. Since addition of real numbers is associative, $(a_{ij}+b_{ij})+c_{ij} = a_{ij}+(b_{ij}+c_{ij})$ for all i, j. Hence $(A+B)+C = A+(B+C)$.
3. *Additive identity* $A + 0 = A$: The (i, j) entry of $A + 0$ is $a_{ij} + 0 = a_{ij}$, which is the (i, j) entry of A. Hence $A + 0 = A$.
4. *Scalar distributivity over matrix sum* $a(A+B) = aA+aB$: The (i, j) entry of $a(A+B)$ is $a(a_{ij} + b_{ij})$, and the (i, j) entry of $aA + aB$ is $aa_{ij} + ab_{ij}$. Since real multiplication distributes over real addition, $a(a_{ij} + b_{ij}) = aa_{ij} + ab_{ij}$ for all i, j. Hence $a(A + B) = aA + aB$.
5. *Scalar distributivity over scalar sum* $(a + b)A = aA + bA$: The (i, j) entry of $(a + b)A$ is $(a + b)a_{ij}$, and the (i, j) entry of $aA + bA$ is $aa_{ij} + ba_{ij}$. Since real multiplication distributes over real addition, $(a + b)a_{ij} = aa_{ij} + ba_{ij}$ for all i, j. Hence $(a + b)A = aA + bA$.
6. *Scalar associativity* $a(bA) = (ab)A$: The (i, j) entry of $a(bA)$ is $a(ba_{ij})$, and the (i, j) entry of $(ab)A$ is $(ab)a_{ij}$. By associativity of real multiplication, $a(ba_{ij}) = (ab)a_{ij}$ for all i, j. Hence $a(bA) = (ab)A$.

■

3.4 Composition and Matrix Multiplication

Definition. If $U : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^k$, the *composition* $T \circ U : \mathbb{R}^n \to \mathbb{R}^k$ is $(T \circ U)(\vec{v}) = T(U(\vec{v}))$. If T and U are linear, then $T \circ U$ is linear.

Order matters

$U \circ T$ is not defined unless $k = n$. Even when both $T \circ U$ and $U \circ T$ are defined, generally $T \circ U \neq U \circ T$ (composition is not commutative).

Definition. If A is $k \times m$ (standard matrix of T) and B is $m \times n$ (standard matrix of U), then the *matrix product* AB is the $k \times n$ standard matrix of $T \circ U$:

$$AB = C \quad \text{where} \quad C\vec{v} = A(B\vec{v}) \text{ for all } \vec{v} \in \mathbb{R}^n.$$

Column-wise formula for AB

Theorem. If $B = \begin{pmatrix} \vec{b}_1 & \vec{b}_2 & \cdots & \vec{b}_n \end{pmatrix}$ then

$$AB = \begin{pmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_n \end{pmatrix}.$$

Each column of AB is A times the corresponding column of B.

Proof. Let A be $m \times p$ with columns $\vec{a}_1, \ldots, \vec{a}_p$, and $B = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix}$ be $p \times n$. By definition, AB is the matrix of the composite $T \circ U$ where $T(\vec{v}) = A\vec{v}$ and $U(\vec{v}) = B\vec{v}$. The jth column of AB is $(AB)\vec{e}_j$. Now $\vec{e}_j$ is the jth standard basis vector of $\mathbb{R}^n$, so

$B\vec{e}_j = \vec{b}_j$ (the jth column of B). Thus $(AB)\vec{e}_j = A(B\vec{e}_j) = A\vec{b}_j$. So the jth column of AB is $A\vec{b}_j$, giving $AB = \begin{pmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_n \end{pmatrix}$. ■

Entry-wise formula. The entry in row i, column j of AB is:

$$(AB)_{ij} = \sum_{k=1}^{m} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

(Row i of A dotted with column j of B.)

Definition. The $n \times n$ *identity matrix* is

$$I_n = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

(1's on diagonal, 0's elsewhere). It is the standard matrix of $\mathrm{id}_{\mathbb{R}^n}(\vec{v}) = \vec{v}$.

Proposition. If A is $m \times n$ then $I_m A = A = AI_n$.

Proof. For $I_m A$: Let $\vec{a}_1, \ldots, \vec{a}_n$ be the columns of A. By the column-wise formula, the jth column of $I_m A$ is $I_m\vec{a}_j$. Since I_m is the identity, $I_m\vec{a}_j = \vec{a}_j$ (the jth column of A). So $I_m A = \begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n \end{pmatrix} = A$. For AI_n: the jth column of AI_n is $A\vec{e}_j$ where $\vec{e}_j$ is the jth standard basis vector of $\mathbb{R}^n$. By the fact that $A\vec{e}_j$ is the jth column of A, we get $AI_n = \begin{pmatrix} A\vec{e}_1 & \cdots & A\vec{e}_n \end{pmatrix} = \begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n \end{pmatrix} = A$. ■

3.5 Properties of Matrix Multiplication

Proposition (Associativity). If A is $m \times n$, B is $n \times l$, C is $l \times k$, then $A(BC) = (AB)C$.

Proof (via function composition). Let $T_A(\vec{v}) = A\vec{v}$, $T_B(\vec{v}) = B\vec{v}$, $T_C(\vec{v}) = C\vec{v}$ be the corresponding linear transformations. The matrix BC represents $T_B \circ T_C$ and AB represents $T_A \circ T_B$. Then $A(BC)$ represents $T_A \circ (T_B \circ T_C)$, and $(AB)C$ represents $(T_A \circ T_B) \circ T_C$. But function composition is associative: $T_A \circ (T_B \circ T_C) = (T_A \circ T_B) \circ T_C$, since both evaluate to $T_A(T_B(T_C(\vec{v})))$ for any $\vec{v}$. Since the two products represent the same linear transformation and the standard matrix is unique, $A(BC) = (AB)C$. ■

Proposition (Distributivity). For matrices of compatible sizes and $r \in \mathbb{R}$:

- $A(B + C) = AB + AC$.
- $(A + B)C = AC + BC$.
- $r(AB) = (rA)B = A(rB)$.

Proof. We prove each. Let A be $m \times n$.

1. $A(B + C) = AB + AC$: Let $\vec{c}_1, \ldots, \vec{c}_k$ be the columns of C and $\vec{b}_1, \ldots, \vec{b}_k$ the columns of B. The jth column of $B+C$ is $\vec{b}_j + \vec{c}_j$, so the jth column of $A(B+C)$ is $A(\vec{b}_j + \vec{c}_j) = A\vec{b}_j + A\vec{c}_j$ (using linearity of matrix-vector product). This equals the jth column of AB plus the jth column of AC, i.e., the jth column of $AB+AC$. So $A(B+C) = AB+AC$.
2. $(A + B)C = AC + BC$: The jth column of $(A + B)C$ is $(A + B)\vec{c}_j = A\vec{c}_j + B\vec{c}_j$ (by definition of matrix addition), which is the jth column of AC plus the jth column of BC, i.e., the jth column of $AC + BC$. So $(A + B)C = AC + BC$.
3. $r(AB) = (rA)B = A(rB)$: The jth column of $r(AB)$ is $r(A\vec{b}_j) = (rA)\vec{b}_j$ (linearity), which is the jth column of $(rA)B$. Also $r(A\vec{b}_j) = A(r\vec{b}_j)$ (linearity), which is the jth

column of $A(rB)$ (whose jth column is $r\vec{b}_j$). So $r(AB) = (rA)B = A(rB)$.

■

3.6 Pathologies of Matrix Multiplication

What matrix multiplication does NOT satisfy

1. AB is defined only if #columns of A = #rows of B.
2. Even if AB and BA are both defined, generally $AB \neq BA$.
3. $AB = AC$ does **not** imply $B = C$ (no cancellation).
4. $AB = 0$ can happen even when $A \neq 0$ and $B \neq 0$ (zero divisors).

Definition. Square matrices A and B of the same size *commute* if $AB = BA$.

3.7 Elementary Matrices and Transpose

Definition. An *elementary matrix* is a matrix formed by performing exactly one row operation on the identity matrix I_n. If E is elementary, then EA is obtained from A by doing that row operation.

Row operations = left multiplication by elementary matrices

Every row operation on A corresponds to left-multiplying A by an elementary matrix E.

Definition. The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^T formed by flipping A across the main diagonal (rows become columns).

Properties of transpose:

- $(A^T)^T = A$.
- $(A + B)^T = A^T + B^T$ and $(cA)^T = c(A^T)$.
- $(AB)^T = B^T A^T$ (order reverses).

Proof. We prove each property.

1. $(A^T)^T = A$: The (i, j) entry of A^T is a_{ji}. So the (i, j) entry of $(A^T)^T$ is $(A^T)_{ji} = a_{ij}$. Hence $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$ and $(cA)^T = c(A^T)$: The (i, j) entry of $(A + B)^T$ is $(A + B)_{ji} = a_{ji} + b_{ji}$, which equals $a_{ji} + b_{ji} = (A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$. So $(A + B)^T = A^T + B^T$. Similarly, the (i, j) entry of $(cA)^T$ is $(cA)_{ji} = c\,a_{ji} = c(A^T)_{ij} = (cA^T)_{ij}$, so $(cA)^T = c(A^T)$.
3. $(AB)^T = B^T A^T$: Let A be $m \times n$ and B be $n \times p$. Then AB is $m \times p$, so $(AB)^T$ is $p \times m$. Also B^T is $p \times n$ and A^T is $n \times m$, so $B^T A^T$ is $p \times m$ (sizes match). The (i, j) entry of $(AB)^T$ is $(AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$. The (i, j) entry of $B^T A^T$ is $\sum_{k=1}^{n} (B^T)_{ik}(A^T)_{kj} = \sum_{k=1}^{n} b_{ki} a_{jk} = \sum_{k=1}^{n} a_{jk} b_{ki}$. These are equal, so $(AB)^T = B^T A^T$.

■

3.8 Invertible Matrices

Definition. $f : X \to Y$ is *invertible* (or *bijective*) if f is both onto and one-to-one.

Proposition. f is invertible $\Leftrightarrow$ for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.

Proof. ($\Rightarrow$) Suppose f is invertible (bijective), so f is one-to-one and onto. Since f is onto, for each $b \in Y$ there exists $a \in X$ with $f(a) = b$. Since f is one-to-one, this a is unique. ($\Leftarrow$) Suppose for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$. Then f is onto (every b has a preimage) and one-to-one (if $f(a_1) = f(a_2)$, then both a_1, a_2 are preimages of the same b, so $a_1 = a_2$ by uniqueness). So f is bijective, i.e., invertible. ■

Proposition. f is invertible $\Leftrightarrow$ there exists a unique function $f^{-1} : Y \to X$ (the *inverse*) such that $f \circ f^{-1} = \mathrm{id}_Y$ and $f^{-1} \circ f = \mathrm{id}_X$.

Proof. ($\Rightarrow$) If f is invertible, for each $b \in Y$ define $f^{-1}(b)$ to be the unique $a \in X$ with $f(a) = b$ (exists and unique by the previous proposition). Then $f(f^{-1}(b)) = b$ so $f \circ f^{-1} = \mathrm{id}_Y$, and $f^{-1}(f(a)) = a$ so $f^{-1} \circ f = \mathrm{id}_X$. For uniqueness: if $g : Y \to X$ satisfies $f \circ g = \mathrm{id}_Y$, then for any $b \in Y$, $f(g(b)) = b$, so $g(b)$ is a preimage of b under f, which is unique; hence $g(b) = f^{-1}(b)$. So f^{-1} is unique. ($\Leftarrow$) If such f^{-1} exists, then for each $b \in Y$, $f(f^{-1}(b)) = b$ shows f is onto, and if $f(a_1) = f(a_2)$, then applying f^{-1} gives $a_1 = f^{-1}(f(a_1)) = f^{-1}(f(a_2)) = a_2$, so f is one-to-one. Thus f is invertible. ■

Proposition. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear and invertible, then $n = m$ and T^{-1} is linear.

Proof. Since T is invertible, it is one-to-one and onto. The fact T is onto requires $n \geq m$ (a linear $T : \mathbb{R}^n \to \mathbb{R}^m$ can be onto only if $n \geq m$, since the columns of the standard matrix must span $\mathbb{R}^m$, requiring at least m columns). The fact T is one-to-one requires $n \leq m$ (the columns of the standard matrix must be linearly independent, requiring at most m columns in $\mathbb{R}^m$). Combining, $n = m$. Now we show $T^{-1} : \mathbb{R}^m \to \mathbb{R}^m$ is linear. For $\vec{u}, \vec{v} \in \mathbb{R}^m$, let $\vec{a} = T^{-1}(\vec{u})$ and $\vec{b} = T^{-1}(\vec{v})$, so $T(\vec{a}) = \vec{u}$ and $T(\vec{b}) = \vec{v}$. Then $T(\vec{a}+\vec{b}) = T(\vec{a}) + T(\vec{b}) = \vec{u}+\vec{v}$ (linearity of T), so $T^{-1}(\vec{u}+\vec{v}) = \vec{a}+\vec{b} = T^{-1}(\vec{u}) + T^{-1}(\vec{v})$. Similarly, $T(c\vec{a}) = cT(\vec{a}) = c\vec{u}$, so $T^{-1}(c\vec{u}) = c\vec{a} = cT^{-1}(\vec{u})$. Hence T^{-1} satisfies both linearity conditions. ■

Definition. An $n \times n$ matrix A is *invertible* (or *non-singular*) if the linear function $T(\vec{x}) = A\vec{x}$ is invertible. The *inverse* of A is the unique matrix A^{-1} such that $T^{-1}(\vec{x}) = A^{-1}\vec{x}$.

Equivalent conditions for invertibility

Proposition. Let A be $n \times n$. The following are equivalent:
1. A is invertible.
2. There exists A^{-1} with $AA^{-1} = A^{-1}A = I_n$.
3. For each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution.
4. $\mathrm{RREF}(A) = I_n$.

Proof. We show (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (4).

(1) $\Leftrightarrow$ (2): (1) says $T(\vec{x}) = A\vec{x}$ is invertible as a function. By the proposition on invertible functions, this is equivalent to the existence of T^{-1} with $T \circ T^{-1} = T^{-1} \circ T = \mathrm{id}$. Since T^{-1} is linear (by the previous proposition) with standard matrix A^{-1}, this means $AA^{-1} = A^{-1}A = I_n$, which is (2).

(2) $\Leftrightarrow$ (3): If (2), then for any $\vec{b}$, $\vec{x} = A^{-1}\vec{b}$ satisfies $A\vec{x} = A(A^{-1}\vec{b}) = (AA^{-1})\vec{b} = I_n\vec{b} = \vec{b}$, so a solution exists. For uniqueness, if $A\vec{x} = \vec{b}$, then multiplying by A^{-1} on the left gives $\vec{x} = A^{-1}\vec{b}$. So the solution is unique. Conversely, if (3), then $A\vec{x} = \vec{e}_j$ has a unique solution

$\vec{c}_j$ for each j. Let $C = (\vec{c}_1 \ \cdots \ \vec{c}_n)$. Then $AC = (A\vec{c}_1 \ \cdots \ A\vec{c}_n) = (\vec{e}_1 \ \cdots \ \vec{e}_n) = I_n$. So $AC = I_n$, and (for square matrices) this gives $C = A^{-1}$, i.e., $AA^{-1} = I_n$.

(3) $\Leftrightarrow$ (4): $A\vec{x} = \vec{b}$ has a unique solution for each $\vec{b}$ iff the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ has a unique solution for each $\vec{b}$. By the RREF criterion, this requires no free variables (every column of A is a pivot column) and the last column never being a pivot. No free variables means every column of A is a pivot column, i.e., RREF(A) has n pivots. Since A is $n \times n$ with n pivots (one in each column, hence one in each row), RREF$(A) = I_n$. This is (4).

■

2×2 inverse formula

Theorem. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

1. A is invertible if and only if $ad - bc \neq 0$.
2. If $ad - bc \neq 0$, then $A^{-1} = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Proof. We first verify (2) by direct computation, assuming $ad - bc \neq 0$. Let $B = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Then

$$AB = \frac{1}{ad-bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{pmatrix}$$
$$= \frac{1}{ad-bc} \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

Similarly, $BA = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \dfrac{1}{ad-bc} \begin{pmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{pmatrix} = I_2$. So $B = A^{-1}$, confirming (2). For (1): If $ad - bc \neq 0$, the computation above gives an explicit A^{-1}, so A is invertible. Conversely, suppose A is invertible, so there exists A^{-1} with $AA^{-1} = I_2$. Then $\det(AA^{-1}) = \det(I_2) = 1$, and if we know $\det(AB) = (\det A)(\det B)$, we get $(\det A)(\det A^{-1}) = 1$, so $\det A \neq 0$. Alternatively, a direct proof: if $ad - bc = 0$, then the columns $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$ satisfy $a \cdot d - b \cdot c = 0$, meaning they are linearly dependent (one is a scalar multiple of the other, or both are zero). If the columns are linearly dependent, then $A\vec{x} = \vec{0}$ has a nontrivial solution, so $T(\vec{x}) = A\vec{x}$ is not one-to-one, hence not invertible. So A invertible implies $ad - bc \neq 0$. ■

Recipe for 2×2 inverse

1. Swap diagonal entries.
2. Negate off-diagonal entries.
3. Divide by $ad - bc$.

Theorem. Let A and B be $n \times n$ matrices.

1. If A is invertible, then $(A^{-1})^{-1} = A$.
2. If A and B are invertible, then AB is invertible with $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is invertible, then A^T is invertible with $(A^T)^{-1} = (A^{-1})^T$.

Proof. We verify each by checking the defining property of the inverse (i.e., finding a matrix C with $AC = CA = I_n$).

1. Since A is invertible, $AA^{-1} = A^{-1}A = I_n$. Reading this as "(the inverse of A^{-1}) times A^{-1} equals I" and "A^{-1} times (the inverse of A^{-1}) equals I", we see A satisfies the

property of being the inverse of A^{-1}. By uniqueness of the inverse, $(A^{-1})^{-1} = A$.

2. Compute:
$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A\,I_n\,A^{-1} = AA^{-1} = I_n,$$
$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$
So $B^{-1}A^{-1}$ serves as the inverse of AB. By uniqueness, $(AB)^{-1} = B^{-1}A^{-1}$.

3. We use $(XY)^T = Y^TX^T$. Take $X = A$ and $Y = A^{-1}$: $(AA^{-1})^T = (A^{-1})^TA^T$. But $AA^{-1} = I_n$, so $(I_n)^T = I_n = (A^{-1})^TA^T$. Similarly, $(A^{-1}A)^T = A^T(A^{-1})^T = I_n$. So $(A^{-1})^T$ serves as the inverse of A^T. By uniqueness, $(A^T)^{-1} = (A^{-1})^T$.

■

General formula for products

$(ABC\cdots Z)^{-1} = Z^{-1}\cdots C^{-1}B^{-1}A^{-1}$ (inverses in reverse order).

3.9 Computing A^{-1}

Algorithm for A^{-1}

Form the $n \times 2n$ matrix $(A \quad I_n)$ and row reduce to reduced echelon form. If A is invertible, then
$$\text{RREF}\left((A \quad I_n)\right) = \left(I_n \quad A^{-1}\right).$$
Take the last n columns as A^{-1}. If $\text{RREF}(A) \neq I_n$, then A is not invertible.

Only square matrices can be invertible

If A is not square, it is possible to have $AB = I$ for some B, but A is still not invertible (there is no A' with both $A'A = I$ and $AA' = I$). For square matrices, $AB = I_n$ suffices to conclude $B = A^{-1}$.

4 Linear Transformations

4.1 Linear Independence

Definition. A list of p vectors $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ is *linearly independent* if
$$\vec{v}_i \notin \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\} \quad \text{for all } i = 1, \ldots, p.$$

Special cases:

- Empty list: always linearly independent (by convention).
- One vector $\vec{v}_1$: independent iff $\vec{v}_1 \neq \vec{0}$.
- Two vectors $\vec{v}_1, \vec{v}_2$: independent iff neither is a scalar multiple of the other.
- Three vectors: independent iff no one is in the span of the other two.

Equivalent characterization

Proposition. $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent if and only if the vector equation
$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$
has **no other solutions** besides the trivial solution $x_1 = x_2 = \cdots = x_p = 0$. A nontrivial solution is called a *linear dependence* among the vectors.

Proof. We prove both directions.

($\Rightarrow$) Suppose $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent. By definition, $\vec{v}_i \notin \text{span}\{\vec{v}_j\}_{j \neq i}$ for all i. Suppose for contradiction that $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ has a nontrivial solution, i.e., some $x_i \neq 0$. Then

$$\vec{v}_i = -\sum_{j \neq i} \frac{x_j}{x_i}\vec{v}_j,$$

which expresses $\vec{v}_i$ as a linear combination of the other vectors. This means $\vec{v}_i \in \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\}$, contradicting the linear independence. So only the trivial solution exists. ■

($\Leftarrow$) Suppose the only solution to $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ is the trivial one. We show each $\vec{v}_i \notin \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\}$. Suppose for contradiction $\vec{v}_i = \sum_{j \neq i} a_j\vec{v}_j$ for some scalars a_j. Then setting $x_i = -1$ and $x_j = a_j$ for $j \neq i$ gives

$$\begin{aligned} x_1\vec{v}_1 + \cdots + x_p\vec{v}_p &= -\vec{v}_i + \sum_{j \neq i} a_j\vec{v}_j \\ &= -\vec{v}_i + \vec{v}_i = \vec{0}, \end{aligned}$$

a nontrivial solution (since $x_i = -1 \neq 0$). Contradiction. So the list is linearly independent.

■

Detecting linear independence

Corollary. The columns of a matrix A are linearly independent if and only if A has a pivot position in **every column**.

Proof. Let $A = (\vec{v}_1 \ \cdots \ \vec{v}_p)$. By the proposition, the columns are linearly independent iff $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ has only the trivial solution, i.e., $A\vec{x} = \vec{0}$ has only the trivial solution $\vec{x} = \vec{0}$. By the RREF criterion, this homogeneous system has a unique solution iff there are no free variables, i.e., every column of A is a pivot column. So the columns of A are linearly independent iff A has a pivot in every column. ■

Algorithm: Form the matrix $A = (\vec{v}_1 \ \ \vec{v}_2 \ \ \cdots \ \ \vec{v}_p)$, reduce to echelon form, and check if every column has a pivot.

4.2 Linear Dependence

Definition. A list is *linearly dependent* if it is not linearly independent.

Sufficient conditions for dependence

Proposition. If $p > n$ then $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ are linearly dependent (more vectors than dimensions).

Proof. Form the matrix $A = (\vec{v}_1 \ \cdots \ \vec{v}_p)$ of size $n \times p$. Since $p > n$, A has more columns than rows. In any echelon form, each pivot is in a distinct row, so there are at most n pivots. Since there are $p > n$ columns, at least one column is not a pivot column, i.e., there is a free variable. So $A\vec{x} = \vec{0}$ has a nontrivial solution, meaning the columns are linearly dependent. ■

Proposition. If any $\vec{v}_i = \vec{0}$ then the list is linearly dependent.

Proof. Suppose $\vec{v}_i = \vec{0}$ for some i. Set $x_i = 1$ and $x_j = 0$ for $j \neq i$. Then $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{v}_i = \vec{0}$, a nontrivial solution (since $x_i = 1 \neq 0$) to $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$. By the equivalent characterization, the list is linearly dependent. ■

4.3 Linear Transformations

Definition. A function $f : X \to Y$ takes inputs from the *domain* X and produces outputs in the *codomain* Y. Not every element of Y needs to be an output.

Definition. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a function. T is a *linear transformation* if:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v} \in \mathbb{R}^n$.
2. $T(c\vec{v}) = cT(\vec{v})$ for all $\vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Proposition. If T is linear, then:

- $T(\vec{0}) = \vec{0}$.
- $T(\vec{u} - \vec{v}) = T(\vec{u}) - T(\vec{v})$.
- $T(a_1\vec{v}_1 + \cdots + a_p\vec{v}_p) = a_1T(\vec{v}_1) + \cdots + a_pT(\vec{v}_p)$ (compatible with linear combinations).

Proof. We prove each property using the two linearity axioms: $T(\vec{u}+\vec{v}) = T(\vec{u})+T(\vec{v})$ and $T(c\vec{v}) = cT(\vec{v})$.

1. $T(\vec{0}) = T(\vec{0} + \vec{0}) = T(\vec{0}) + T(\vec{0})$ (using additivity). Subtracting $T(\vec{0})$ from both sides gives $\vec{0} = T(\vec{0})$.
2. $T(\vec{u} - \vec{v}) = T(\vec{u} + (-1)\vec{v}) = T(\vec{u}) + T((-1)\vec{v}) = T(\vec{u}) + (-1)T(\vec{v}) = T(\vec{u}) - T(\vec{v})$ (using additivity and homogeneity).
3. By induction on p. Base case $p = 1$: $T(a_1\vec{v}_1) = a_1T(\vec{v}_1)$ (homogeneity). Inductive step: suppose it holds for $p - 1$ vectors. Then $T(a_1\vec{v}_1 + \cdots + a_p\vec{v}_p) = T\big((a_1\vec{v}_1 + \cdots + a_{p-1}\vec{v}_{p-1}) + a_p\vec{v}_p\big) = T(a_1\vec{v}_1 + \cdots + a_{p-1}\vec{v}_{p-1}) + T(a_p\vec{v}_p)$ (additivity). By induction, this is $a_1T(\vec{v}_1) + \cdots + a_{p-1}T(\vec{v}_{p-1}) + a_pT(\vec{v}_p)$. Done.

■

Definition. The *standard basis* vectors of $\mathbb{R}^n$ are $\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, \vec{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$.

Fact. If A is $m \times n$, then $A\vec{e}_i$ is the ith column of A.

Proof: Let $A = (\vec{a}_1 \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n)$ where $\vec{a}_j \in \mathbb{R}^m$ is the jth column. The standard basis vector $\vec{e}_i \in \mathbb{R}^n$ has ith entry equal to 1 and all other entries equal to 0. By the definition of the matrix-vector product,

$$A\vec{e}_i = (\vec{a}_1 \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = 0 \cdot \vec{a}_1 + \cdots + 1 \cdot \vec{a}_i + \cdots + 0 \cdot \vec{a}_n = \vec{a}_i.$$

So $A\vec{e}_i = \vec{a}_i$, the ith column of A. ■

Fundamental theorem of linear transformations

Theorem. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then there is a **unique** $m \times n$ matrix A such that $T(\vec{v}) = A\vec{v}$ for all $\vec{v} \in \mathbb{R}^n$. This matrix is

$$A = \begin{pmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{pmatrix}.$$

We call A the *standard matrix* of T.

Proof. **Existence:** Define $A = \begin{pmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{pmatrix}$, an $m \times n$ matrix (each $T(\vec{e}_i) \in \mathbb{R}^m$). For any $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$, we have $\vec{v} = v_1\vec{e}_1 + v_2\vec{e}_2 + \cdots + v_n\vec{e}_n$ (this is how $\vec{v}$ decomposes in the standard basis). By linearity of T (repeated use of additivity and homogeneity, i.e., the previous proposition), $T(\vec{v}) = T(v_1\vec{e}_1 + \cdots + v_n\vec{e}_n) = v_1T(\vec{e}_1) + \cdots + v_nT(\vec{e}_n)$. By the definition of matrix-vector product, $v_1T(\vec{e}_1) + \cdots + v_nT(\vec{e}_n) = \begin{pmatrix} T(\vec{e}_1) & \cdots & T(\vec{e}_n) \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = A\vec{v}$. So $T(\vec{v}) = A\vec{v}$ for all $\vec{v}$, as needed. **Uniqueness:** Suppose B is another $m \times n$ matrix with $T(\vec{v}) = B\vec{v}$ for all $\vec{v} \in \mathbb{R}^n$. Then for each standard basis vector $\vec{e}_i$, $T(\vec{e}_i) = B\vec{e}_i$. But $B\vec{e}_i$ is the ith column of B. So the ith column of B equals $T(\vec{e}_i)$, which is the ith column of A. Hence $A = B$, proving uniqueness. ■

Not every function is linear

$T(\vec{v}) = v_1^2 + \cdots + v_n^2$ is **not** linear since $T(2\vec{v}) = 4T(\vec{v}) \neq 2T(\vec{v})$. Functions of the form $f(x) = ax + b$ with $b \neq 0$ are **not** linear (they are "affine").

4.4 Functions and Range

Definition. For $f : X \to Y$:

- The *image* of x under f is $f(x)$.
- The *range* (or *image*) of f is $\text{range}(f) = \{f(x) : x \in X\} \subseteq Y$.

Two functions are the same only when domain, codomain, and formula are all equal.

4.5 One-to-one (Injective)

Definition. $f : X \to Y$ is *one-to-one* (or *injective*) if $f(a) = f(b)$ implies $a = b$. Equivalently, f does not send two different inputs to the same output.

Detecting one-to-one for linear T

Theorem. Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear with standard matrix A ($m \times n$). The following are equivalent:

(a) T is one-to-one.

(b) The only solution to $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$.

(c) The columns of A are linearly independent.

(d) Every column of A contains a pivot position.

Proof. We show (a) ⇔ (b) ⇔ (c) ⇔ (d).

(a) $\Leftrightarrow$ (b): ($\Rightarrow$) If T is one-to-one and $T(\vec{x}) = \vec{0}$, since $T(\vec{0}) = \vec{0}$ (by linearity), we have $T(\vec{x}) = T(\vec{0})$, so one-to-one gives $\vec{x} = \vec{0}$. ($\Leftarrow$) Suppose the only solution to $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$. If $T(\vec{x}) = T(\vec{y})$, then by linearity $T(\vec{x} - \vec{y}) = T(\vec{x}) - T(\vec{y}) = \vec{0}$. By assumption, $\vec{x} - \vec{y} = \vec{0}$, so $\vec{x} = \vec{y}$. Thus T is one-to-one.

(b) $\Leftrightarrow$ (c): $T(\vec{x}) = \vec{0}$ is $A\vec{x} = \vec{0}$, i.e., $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{0}$. By the linear independence characterization, this has only the trivial solution iff the columns $\vec{a}_1, \ldots, \vec{a}_n$ are linearly independent. So (b) $\Leftrightarrow$ (c).

(c) $\Leftrightarrow$ (d): By the corollary in Section 4 (detecting linear independence), the columns of A are linearly independent iff A has a pivot in every column. So (c) $\Leftrightarrow$ (d).

■

4.6 Onto (Surjective)

Definition. $f : X \to Y$ is *onto* (or *surjective*) if range$(f) = Y$. Equivalently, for each $y \in Y$ there exists $x \in X$ with $f(x) = y$.

Detecting onto for linear T

Theorem. Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear with standard matrix A ($m \times n$). The following are equivalent:

(a) T is onto.

(b) $A\vec{x} = \vec{b}$ has a solution for each $\vec{b} \in \mathbb{R}^m$.

(c) The span of the columns of A is $\mathbb{R}^m$.

(d) Every row of A contains a pivot position.

Proof: (a) $\Leftrightarrow$ (b): By definition, T is onto means range$(T) = \mathbb{R}^m$. But range$(T) = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$. So T is onto iff for every $\vec{b} \in \mathbb{R}^m$ there exists $\vec{x} \in \mathbb{R}^n$ with $A\vec{x} = \vec{b}$, which is exactly (b).

(b) $\Leftrightarrow$ (c): The span of the columns of A is $\{x_1\vec{a}_1 + \cdots + x_n\vec{a}_n : x_i \in \mathbb{R}\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$. So (c) says $\{A\vec{x} : \vec{x} \in \mathbb{R}^n\} = \mathbb{R}^m$, which is exactly (b).

(b) $\Leftrightarrow$ (d): The system $A\vec{x} = \vec{b}$ is consistent iff the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ has no row of the form $(0 \ \cdots \ 0 \mid b)$ with $b \neq 0$ after row reduction. This is equivalent to: every row of A has a pivot position (since if a row of A has no pivot, then for some $\vec{b}$ — namely a vector with a nonzero entry in that position — the augmented column becomes a pivot and the system is inconsistent). Thus $A\vec{x} = \vec{b}$ is solvable for **every** $\vec{b}$ iff every row of A has a pivot, which is (d). ■

4.7 Size Constraints

Dimension constraints

Corollary. For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$:

- T can be one-to-one only if $n \leq m$ (domain no larger than codomain).
- T can be onto only if $n \geq m$ (domain no smaller than codomain).

Proof: For one-to-one: If T is one-to-one, then every column of the standard matrix A has a pivot (by the one-to-one theorem). Since A is $m \times n$ and each of the n columns has a pivot, we need at least n rows to hold these pivots, so $n \leq m$.

For onto: If T is onto, then every row of A has a pivot (by the onto theorem). Since A is $m \times n$ and each of the m rows has a pivot, we need at least m columns to hold these pivots, so $n \geq m$. ■

One-to-one vs. onto depends on different things

Whether T is one-to-one depends on domain and formula, but **not** on the codomain. Whether T is onto **does** depend on the choice of codomain.

5 Subspaces, Basis, and Dimension

5.1 Invertibility: Summary for Square Matrices

Full characterization (square matrices)

Theorem. When A is $n \times n$, the following are equivalent:
1. A is invertible.
2. The columns of A are linearly independent.
3. The span of the columns of A is $\mathbb{R}^n$.

Corollary. If A and B are both $n \times n$ and $AB = I_n$, then $BA = I_n$. (For square matrices, checking $AB = I$ suffices.)

Proof of the theorem: (1) $\Rightarrow$ (2): Suppose A is invertible with inverse A^{-1}. To show the columns are linearly independent, suppose $A\vec{x} = \vec{0}$. Multiplying both sides by A^{-1} on the left gives $\vec{x} = A^{-1}\vec{0} = \vec{0}$. Thus the only solution to $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$, so the columns are linearly independent.

(2) $\Rightarrow$ (3): Suppose the columns are linearly independent. Since A is $n \times n$, the n columns are linearly independent in $\mathbb{R}^n$. For a list of n linearly independent vectors in $\mathbb{R}^n$ (which has dimension n), they must form a basis, so their span is $\mathbb{R}^n$. (Equivalently, since the columns are independent, every column is a pivot column, so $\operatorname{rank} A = n$, meaning $\dim \operatorname{Col} A = n$, and since $\operatorname{Col} A \subseteq \mathbb{R}^n$ has the same dimension, $\operatorname{Col} A = \mathbb{R}^n$.)

(3) $\Rightarrow$ (1): Suppose the columns span $\mathbb{R}^n$. Then for each standard basis vector $\vec{e}_j$, there exists $\vec{b}_j$ with $A\vec{b}_j = \vec{e}_j$. Let $B = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix}$. Then $AB = \begin{pmatrix} A\vec{b}_1 & \cdots & A\vec{b}_n \end{pmatrix} = \begin{pmatrix} \vec{e}_1 & \cdots & \vec{e}_n \end{pmatrix} = I_n$. By the corollary below (whose proof does not depend on this theorem), $BA = I_n$ as well, so A is invertible. ■

Proof of the corollary: Suppose $AB = I_n$. We show A is invertible, which then gives $BA = I_n$. Since $AB = I_n$, for each $\vec{b} \in \mathbb{R}^n$ we have $A(B\vec{b}) = (AB)\vec{b} = I_n\vec{b} = \vec{b}$, so every $\vec{b} \in \mathbb{R}^n$ is in the span of the columns of A. By (3) $\Rightarrow$ (1) of the full characterization (which requires only the fact that if columns span then $A\vec{x} = \vec{e}_j$ is solvable, giving a right inverse), A is invertible. Then $BA = A^{-1}(AB)A = A^{-1}I_nA = I_n$. More directly: $AB = I_n$ and A invertible give $B = A^{-1}$, so $BA = A^{-1}A = I_n$.

(To avoid circular reasoning: from $AB = I_n$, the columns of A span $\mathbb{R}^n$ as shown above. Since A is $n \times n$ with columns spanning $\mathbb{R}^n$ (dimension n), and there are exactly n columns, they must also be linearly independent — otherwise a dependent list of n vectors would span a space of dimension $< n$. Hence A is invertible by the full characterization (2) $\Leftrightarrow$ (1), and $B = A^{-1}$.) ■

5.2 Subspaces of $\mathbb{R}^n$

Definition. A subset $H \subseteq \mathbb{R}^n$ is a *subspace* if:

1. $\vec{0} \in H$.
2. $\vec{u} + \vec{v} \in H$ for all $\vec{u}, \vec{v} \in H$ (closed under addition).
3. $c\vec{v} \in H$ for all $c \in \mathbb{R}$, $\vec{v} \in H$ (closed under scalar multiplication).

Equivalently, a subspace is a nonempty subset closed under linear combinations.

Examples of subspaces

- $\mathbb{R}^n$ itself.
- $\{\vec{0}\}$ (the zero subspace).
- The span of any set of vectors in $\mathbb{R}^n$.
- The range of a linear function $T : \mathbb{R}^k \to \mathbb{R}^n$.
- $\{\vec{v} : T(\vec{v}) = \vec{0}\}$ for a linear $T : \mathbb{R}^n \to \mathbb{R}^k$.

Subspaces of $\mathbb{R}^2$: $\{\vec{0}\}$, lines through origin, or $\mathbb{R}^2$. Subspaces of $\mathbb{R}^3$: $\{\vec{0}\}$, lines through origin, planes through origin, or $\mathbb{R}^3$.

Proof that $\text{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ is a subspace: Let $H = \text{span}\{\vec{v}_1, \ldots, \vec{v}_p\} \subseteq \mathbb{R}^n$. We verify the three subspace conditions.

(i) $\vec{0} \in H$: Write $\vec{0} = 0 \cdot \vec{v}_1 + 0 \cdot \vec{v}_2 + \cdots + 0 \cdot \vec{v}_p$, which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with all coefficients 0). So $\vec{0} \in H$.

(ii) Closed under addition: Let $\vec{u}, \vec{w} \in H$. Then $\vec{u} = a_1\vec{v}_1 + \cdots + a_p\vec{v}_p$ and $\vec{w} = b_1\vec{v}_1 + \cdots + b_p\vec{v}_p$ for some scalars $a_i, b_i \in \mathbb{R}$. Then

$$\vec{u} + \vec{w} = (a_1 + b_1)\vec{v}_1 + \cdots + (a_p + b_p)\vec{v}_p,$$

which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with coefficients $a_i + b_i$). So $\vec{u} + \vec{w} \in H$.

(iii) Closed under scalar multiplication: Let $\vec{u} \in H$ and $c \in \mathbb{R}$. Write $\vec{u} = a_1\vec{v}_1 + \cdots + a_p\vec{v}_p$. Then

$$c\vec{u} = (ca_1)\vec{v}_1 + \cdots + (ca_p)\vec{v}_p,$$

which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with coefficients ca_i). So $c\vec{u} \in H$.

Hence $H = \text{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ is a subspace of $\mathbb{R}^n$. ■

Proof that the range of a linear function is a subspace: Let $T : \mathbb{R}^k \to \mathbb{R}^n$ be linear, and let $H = \text{range}(T) = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^k\} \subseteq \mathbb{R}^n$. We verify the three subspace conditions.

(i) $\vec{0} \in H$: Since T is linear, $T(\vec{0}) = \vec{0}$ (proven earlier). So $\vec{0} = T(\vec{0}) \in \text{range}(T) = H$.

(ii) Closed under addition: Let $\vec{u}, \vec{w} \in H$. Then $\vec{u} = T(\vec{x})$ and $\vec{w} = T(\vec{y})$ for some $\vec{x}, \vec{y} \in \mathbb{R}^k$. By linearity of T, $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y}) = \vec{u} + \vec{w}$. So $\vec{u} + \vec{w} = T(\vec{x} + \vec{y}) \in H$ (since $\vec{x} + \vec{y} \in \mathbb{R}^k$).

(iii) Closed under scalar multiplication: Let $\vec{u} \in H$ and $c \in \mathbb{R}$. Write $\vec{u} = T(\vec{x})$ for some $\vec{x} \in \mathbb{R}^k$. By linearity of T, $T(c\vec{x}) = cT(\vec{x}) = c\vec{u}$. So $c\vec{u} = T(c\vec{x}) \in H$ (since $c\vec{x} \in \mathbb{R}^k$).

Hence $\text{range}(T)$ is a subspace of $\mathbb{R}^n$. ■

Not a subspace

The set $\{\vec{v} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 1\}$ is **not** a subspace (does not contain $\vec{0}$). But

$\{\vec{v} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0\}$ **is** a subspace.

5.3 Column Space and Null Space

Definition. The *column space* of an $m \times n$ matrix A is

$$\operatorname{Col} A = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m.$$

This is the span of the columns of A.

Definition. The *null space* of an $m \times n$ matrix A is

$$\operatorname{Nul} A = \{\vec{v} \in \mathbb{R}^n : A\vec{v} = \vec{0}\} \subseteq \mathbb{R}^n.$$

This is the set of solutions to $A\vec{x} = \vec{0}$.

Key facts

- $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$ (m = #rows of A).
- $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$ (n = #columns of A).
- If $T(\vec{x}) = A\vec{x}$, then $\operatorname{Col} A = \operatorname{range}(T)$ and $\operatorname{Nul} A = \{\vec{x} : T(\vec{x}) = \vec{0}\}$.
- $\vec{b} \in \operatorname{Col} A \Leftrightarrow A\vec{x} = \vec{b}$ has a solution.
- $\operatorname{Col} A = \mathbb{R}^m \Leftrightarrow A\vec{x} = \vec{b}$ has a solution for each $\vec{b}$.

Proof that $\operatorname{Nul} A$ **is a subspace:** We verify the three subspace conditions. (i) $\vec{0} \in \operatorname{Nul} A$: by linearity of matrix multiplication, $A\vec{0} = \vec{0}$, so $\vec{0} \in \operatorname{Nul} A$.

(ii) Closed under addition: Let $\vec{u}, \vec{v} \in \operatorname{Nul} A$, so $A\vec{u} = \vec{0}$ and $A\vec{v} = \vec{0}$. By linearity of matrix multiplication, $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0}$, so $\vec{u} + \vec{v} \in \operatorname{Nul} A$.

(iii) Closed under scalar multiplication: Let $\vec{v} \in \operatorname{Nul} A$ and $c \in \mathbb{R}$. Then $A(c\vec{v}) = c(A\vec{v}) = c\vec{0} = \vec{0}$, so $c\vec{v} \in \operatorname{Nul} A$.

Hence $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$. ■

Proof that $\operatorname{Col} A$ **is a subspace:** Let A be an $m \times n$ matrix. Since $\operatorname{Col} A = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} = \operatorname{range}(T)$ where $T(\vec{x}) = A\vec{x}$ is a linear transformation, and the range of any linear transformation is a subspace (proven above), $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Alternatively, since $\operatorname{Col} A = \operatorname{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ (the span of the columns of A), and the span of any set of vectors is a subspace (proven above), $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$. ■

5.4 Basis

Definition. A *basis* for a subspace $H \subseteq \mathbb{R}^n$ is a set of vectors $\vec{v}_1, \ldots, \vec{v}_k \in H$ that are linearly independent and whose span is H. The empty set $\emptyset$ is a basis for $\{\vec{0}\}$.

Standard basis

The vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$ form the *standard basis* of $\mathbb{R}^n$. They are linearly independent (since $\operatorname{RREF}(I_n) = I_n$ has pivots in all columns) and span $\mathbb{R}^n$.

Theorem. Every subspace H of $\mathbb{R}^n$ has a basis of size at most n.

Proof: If $H = \{\vec{0}\}$, the empty set is a basis. Otherwise pick any nonzero $\vec{v}_1 \in H$; this is a linearly independent set. If $\operatorname{span}\{\vec{v}_1\} = H$ we are done. Otherwise pick $\vec{v}_2 \in H \setminus$

span$\{\vec{v}_1\}$. Then $\{\vec{v}_1, \vec{v}_2\}$ is linearly independent: if $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{0}$ with $c_2 \neq 0$, then $\vec{v}_2 = -(c_1/c_2)\vec{v}_1 \in \text{span}\{\vec{v}_1\}$, a contradiction. Continuing, if $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is independent and does not span H, pick $\vec{v}_{k+1} \in H \backslash \text{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$; the same argument shows $\{\vec{v}_1, \ldots, \vec{v}_{k+1}\}$ is independent. This process must terminate with $k \leq n$ since any list of more than n vectors in $\mathbb{R}^n$ is linearly dependent (more vectors than dimensions). When it terminates we have an independent set that spans H, i.e., a basis. ■

Finding a Basis for Nul A

1. Row reduce A to find RREF(A) and solve $A\vec{x} = \vec{0}$.
2. Express each basic variable in terms of free variables.
3. Write $\vec{x}$ as a linear combination of vectors $\vec{b}_1, \ldots, \vec{b}_k$ (one per free variable).
4. These vectors form a basis for Nul A.

Why the basis vectors are linearly independent

Look at the rows of the coefficient matrix that correspond to the free variables. Each basis vector has a 1 in the position of one free variable and 0 in the positions of the other free variables, making the set linearly independent.

Finding a Basis for Col A

Pivot columns form a basis

Proposition. The pivot columns of A form a basis for Col A.
Note: use the pivot columns of the **original** matrix A, not the pivot columns of RREF(A).

Proof: Let $\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}$ be the pivot columns of A (in the original matrix), corresponding to the pivot columns $i_1 < \cdots < i_k$ of RREF(A). We must show they span Col A and are linearly independent.

Spanning: Let $\vec{a}_j$ be any non-pivot column of A. In RREF(A), column j is a linear combination of the pivot columns to its left (this is the defining property of RREF — each non-pivot column's entries are determined by the leading 1's in pivot positions in the same rows). Since row operations are linear combinations of rows and preserve linear relations among **columns**, the same relation holds for A: $\vec{a}_j = \sum_{i_l < j} c_l \vec{a}_{i_l}$. Thus every non-pivot column is in span$\{\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}\}$, and so Col $A = \text{span}\{\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}\}$.

Linear independence: Suppose $\sum_l c_l \vec{a}_{i_l} = \vec{0}$. This same relation holds in RREF(A): $\sum_l c_l \vec{r}_{i_l} = \vec{0}$, where $\vec{r}_{i_l}$ are the corresponding pivot columns of RREF(A). But the pivot columns of RREF(A) are standard basis vectors $\vec{e}_1, \ldots, \vec{e}_k$ (in distinct rows), which are linearly independent. Hence all $c_l = 0$.

Thus the pivot columns of A form a basis for Col A. ■

5.5 Operations on Subspaces

Definition. The *sum* of two subspaces U, V is $U + V = \{\vec{u} + \vec{v} : \vec{u} \in U, \vec{v} \in V\}$ (a subspace).

The *intersection* $U \cap V$ is always a subspace.

Union is not a subspace

The union $U \cup V$ of two subspaces is generally **not** a subspace (e.g., two different lines through the origin in $\mathbb{R}^2$).

Properties: If A, B have the same number of rows, $\operatorname{Col}(A \quad B) = \operatorname{Col} A + \operatorname{Col} B$. If A, B have the same number of columns, $\operatorname{Nul}\begin{pmatrix} A \\ B \end{pmatrix} = \operatorname{Nul} A \cap \operatorname{Nul} B$.

Basis for $U + V$ (Span of the Union)

Method: basis for the sum

Since $U + V = \operatorname{span}(U \cup V)$, put the basis vectors of U and of V together as the columns of one matrix M; the pivot columns of the **original** M form a basis for $U + V$.

- *Column spaces*: $\operatorname{Col} A + \operatorname{Col} B = \operatorname{Col}(A \quad B)$; take the pivot columns of the original $(A \quad B)$.
- *Null spaces*: find bases of $\operatorname{Nul} A$ and $\operatorname{Nul} B$ (by solving $A\vec{x} = \vec{0}$ and $B\vec{x} = \vec{0}$), place these basis vectors as the columns of M, and take the pivot columns of the original M.

Proof: $U + V = \{\vec{u} + \vec{v} : \vec{u} \in U, \vec{v} \in V\} = \operatorname{span}(U \cup V)$. If $\{\vec{u}_i\}$ spans U and $\{\vec{v}_j\}$ spans V, then $\operatorname{span}(U \cup V) = \operatorname{span}\{\vec{u}_i, \vec{v}_j\} = \operatorname{Col} M$, and the pivot columns of the original M form a basis for $\operatorname{Col} M$ (proven above). For column spaces, the columns of $(A \quad B)$ are exactly the columns of A together with those of B, whose span is $\operatorname{Col} A + \operatorname{Col} B$. ■

Basis for $\operatorname{Col} A \cap \operatorname{Col} B$

Method: intersection of column spaces

Let A be $m \times n$ and B be $m \times p$.

1. $\vec{w} \in \operatorname{Col} A \cap \operatorname{Col} B \Leftrightarrow \vec{w} = A\vec{x} = B\vec{y}$ for some $\vec{x} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^p$, i.e.,
$$(A \quad -B)\begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \vec{0}.$$
2. Find a basis of $\operatorname{Nul}(A \quad -B)$; write each basis vector as $\begin{pmatrix} \vec{x}_i \\ \vec{y}_i \end{pmatrix}$ with $\vec{x}_i \in \mathbb{R}^n$, $\vec{y}_i \in \mathbb{R}^p$.
3. Compute $\vec{w}_i = A\vec{x}_i$ $(= B\vec{y}_i)$. Then $\{\vec{w}_1, \ldots, \vec{w}_k\}$ **spans** $\operatorname{Col} A \cap \operatorname{Col} B$.
4. The $\vec{w}_i$ need not be independent: row reduce $(\vec{w}_1 \quad \cdots \quad \vec{w}_k)$ and take its pivot columns. They form a basis for $\operatorname{Col} A \cap \operatorname{Col} B$.

Proof: *Step 1:* $\vec{w} = A\vec{x} = B\vec{y} \Leftrightarrow A\vec{x} - B\vec{y} = \vec{0} \Leftrightarrow (A \quad -B)\begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \vec{0}$.

Steps 2–3: By step 1, the intersection equals $\{A\vec{x} : \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} \in \operatorname{Nul}(A \quad -B)\}$. Let $P = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = B\vec{y} \text{ for some } \vec{y}\}$. Every $\vec{x} \in P$ comes from some $\begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \sum_i c_i \begin{pmatrix} \vec{x}_i \\ \vec{y}_i \end{pmatrix}$, and comparing the top blocks gives $\vec{x} = \sum_i c_i \vec{x}_i$, so $\{\vec{x}_i\}$ spans P. Hence the intersection is $\{A\vec{x} : \vec{x} \in P\} = \operatorname{span}\{A\vec{x}_i\} = \operatorname{span}\{\vec{w}_i\}$.

Step 4: The $\vec{w}_i$ may be linearly dependent (e.g., if $\operatorname{Nul} A \neq \{\vec{0}\}$, any $\begin{pmatrix}\vec{x}\\ \vec{0}\end{pmatrix}$ with $\vec{x} \in \operatorname{Nul} A$, $\vec{x} \neq \vec{0}$, gives $\vec{w} = A\vec{x} = \vec{0}$). Discarding non-pivot columns removes only vectors that are linear combinations of the pivot columns, so the span is unchanged and the remaining pivot columns form a basis. ■

Basis for $\operatorname{Nul} A \cap \operatorname{Nul} B$

Method: intersection of null spaces

Stack the matrices: $\operatorname{Nul} A \cap \operatorname{Nul} B = \operatorname{Nul} \begin{pmatrix}A\\ B\end{pmatrix}$. Solve $\begin{pmatrix}A\\ B\end{pmatrix}\vec{x} = \vec{0}$ by the standard null-space method; the resulting vectors form a basis for the intersection.

Proof: $\vec{x} \in \operatorname{Nul} A \cap \operatorname{Nul} B \Leftrightarrow A\vec{x} = \vec{0}$ and $B\vec{x} = \vec{0} \Leftrightarrow \begin{pmatrix}A\\ B\end{pmatrix}\vec{x} = \begin{pmatrix}A\vec{x}\\ B\vec{x}\end{pmatrix} = \vec{0} \Leftrightarrow \vec{x} \in \operatorname{Nul}\begin{pmatrix}A\\ B\end{pmatrix}$. ■

Dimension formula

For any two subspaces $U, V \subseteq \mathbb{R}^n$:

$$\dim(U+V) = \dim U + \dim V - \dim(U \cap V).$$

Proof: Let $\{\vec{w}_1, \ldots, \vec{w}_k\}$ be a basis of $U \cap V$. Extend it to a basis $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{u}_1, \ldots, \vec{u}_p\}$ of U and to a basis $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{v}_1, \ldots, \vec{v}_q\}$ of V (so $\dim U = k+p$ and $\dim V = k+q$). We claim

$$\mathcal{B} = \{\vec{w}_1, \ldots, \vec{w}_k, \vec{u}_1, \ldots, \vec{u}_p, \vec{v}_1, \ldots, \vec{v}_q\}$$

is a basis of $U+V$.

Spanning: any $\vec{u}+\vec{v} \in U+V$ is a combination of the $\vec{w}$'s and $\vec{u}$'s (for $\vec{u}$) plus a combination of the $\vec{w}$'s and $\vec{v}$'s (for $\vec{v}$), hence a combination of the vectors in $\mathcal{B}$.

Independence: suppose

$$\sum_i a_i\vec{w}_i + \sum_j b_j\vec{u}_j + \sum_l c_l\vec{v}_l = \vec{0}.$$

Then $\sum_l c_l\vec{v}_l = -\left(\sum_i a_i\vec{w}_i + \sum_j b_j\vec{u}_j\right) \in U$; being a combination of the $\vec{v}$'s, it is also in V, hence in $U \cap V$. So $\sum_l c_l\vec{v}_l = \sum_i d_i\vec{w}_i$ for some scalars d_i. Since $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{v}_1, \ldots, \vec{v}_q\}$ is a basis of V (linearly independent), all $c_l = 0$. The relation reduces to $\sum_i a_i\vec{w}_i + \sum_j b_j\vec{u}_j = \vec{0}$, and since $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{u}_1, \ldots, \vec{u}_p\}$ is a basis of U, all $a_i = 0$ and all $b_j = 0$. Hence $\mathcal{B}$ is linearly independent.

Therefore $\dim(U+V) = k+p+q = (k+p)+(k+q)-k = \dim U + \dim V - \dim(U \cap V)$. ■

Example. Let $A = \begin{pmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{pmatrix}$ and $B = \begin{pmatrix}1 & 0\\ 1 & 0\\ 0 & 1\end{pmatrix}$.

Sum: $(A \quad B) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ is already in RREF with pivots in columns $1, 2, 4$, so

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ is a basis of } \operatorname{Col} A + \operatorname{Col} B = \mathbb{R}^3.$$

Intersection: row reduce

$$(A \quad -B) = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The free variable is $y_1 = t$, giving $x_1 = t$, $x_2 = t$, $y_2 = 0$, so $\operatorname{Nul}(A \quad -B) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\}$.

The $\vec{x}$-part is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Hence

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\} \text{ is a basis of } \operatorname{Col} A \cap \operatorname{Col} B.$$

Dimension check: $3 = 2 + 2 - 1$, consistent with the dimension formula.

5.6 Coordinate Systems

Theorem. Let H be a subspace with basis $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_k)$. For each $\vec{v} \in H$, there exist **unique** coefficients $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$.

Proof: *Existence:* Since $\mathcal{B}$ is a basis, it spans H. Since $\vec{v} \in H$, there exist $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$.

Uniqueness: Suppose $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k = d_1\vec{b}_1 + \cdots + d_k\vec{b}_k$. Subtracting, $(c_1 - d_1)\vec{b}_1 + \cdots + (c_k - d_k)\vec{b}_k = \vec{0}$. Since $\mathcal{B}$ is a basis, the $\vec{b}_i$ are linearly independent, so $c_i - d_i = 0$ for all i, i.e., $c_i = d_i$. Hence the coefficients are unique. ■

Definition. The *coordinate vector* of $\vec{v}$ in the basis $\mathcal{B}$ is

$$[\vec{v}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} \in \mathbb{R}^k$$

where $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$. Equivalently, $[\vec{v}]_{\mathcal{B}}$ is the unique solution to $\begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_k \end{pmatrix} \vec{x} = \vec{v}$.

The map $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is linear and invertible

The function $H \to \mathbb{R}^k$ given by $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is a linear transformation and is invertible. Thus H "looks the same as" $\mathbb{R}^k$.

5.7 Dimension

All bases have the same size

Theorem. Every basis of a subspace $H \subseteq \mathbb{R}^n$ has the same number of elements.

Proof: Suppose for contradiction that $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_k)$ and $\mathcal{B}' = (\vec{b}'_1, \ldots, \vec{b}'_l)$ are two bases of H with $k < l$.

Consider the coordinate vectors $[\vec{b}'_1]_{\mathcal{B}}, \ldots, [\vec{b}'_l]_{\mathcal{B}}$, which are vectors in $\mathbb{R}^k$. Since $l > k$, this is a list of more than k vectors in $\mathbb{R}^k$, so by the proposition "$p > n \Rightarrow$ linearly dependent" (applied with $n = k$, $p = l$), these vectors are linearly dependent. Thus there exist scalars $c_1, \ldots, c_l$, not all zero, such that

$$c_1[\vec{b}'_1]_{\mathcal{B}} + \cdots + c_l[\vec{b}'_l]_{\mathcal{B}} = \vec{0} \in \mathbb{R}^k.$$

Since the coordinate map $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is linear and invertible (hence injective), this implies

$$c_1\vec{b}'_1 + \cdots + c_l\vec{b}'_l = \vec{0}$$

with not all $c_i = 0$. This contradicts the linear independence of $\mathcal{B}'$. Hence $l \leq k$. By symmetry (exchanging $\mathcal{B}$ and $\mathcal{B}'$), $k \leq l$. Therefore $k = l$. ■

Definition. The *dimension* of a subspace H, denoted $\dim H$, is the number of vectors in any basis of H. We have $\dim \mathbb{R}^n = n$ and $\dim\{\vec{0}\} = 0$.

Basis theorem

Theorem (Basis theorem). If H is a subspace of $\mathbb{R}^n$ with $\dim H = p$, then:
1. Any set of p linearly independent vectors in H is a basis for H.
2. Any set of p vectors in H whose span is H is a basis for H.

Proof: (i) Let $\vec{v}_1, \ldots, \vec{v}_p$ be linearly independent vectors in H, where $p = \dim H$. We show they span H. Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_p)$ be a basis of H (which has p elements). Suppose $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} \neq H$. Then there exists $\vec{v}_{p+1} \in H \setminus \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$. By the standard argument (used in the proof that every subspace has a basis), $\{\vec{v}_1, \ldots, \vec{v}_p, \vec{v}_{p+1}\}$ is linearly independent, giving $p+1$ linearly independent vectors in H. But then $\{[\vec{v}_1]_{\mathcal{B}}, \ldots, [\vec{v}_{p+1}]_{\mathcal{B}}\}$ is a list of $p+1$ vectors in $\mathbb{R}^p$, hence linearly dependent — leading to a nontrivial relation among $\vec{v}_1, \ldots, \vec{v}_{p+1}$ (since the coordinate map is injective), contradicting their independence. So $\vec{v}_1, \ldots, \vec{v}_p$ span H, and being independent they form a basis. ■

(ii) Let $\vec{v}_1, \ldots, \vec{v}_p$ be vectors in H with $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} = H$. Suppose they are linearly dependent. Then some $\vec{v}_j$ is a linear combination of the others, so

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_{j-1}, \vec{v}_{j+1}, \ldots, \vec{v}_p\} = H.$$

This gives a spanning set of H with $p-1$ vectors, from which we can extract a basis of size $\leq p-1$. But every basis of H has size $p = \dim H$ by the previous theorem, a contradiction. Hence $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent, and being a spanning set, they form a basis. ■

5.8 Rank

Definition. The *rank* of a matrix A is $\operatorname{rank} A = \dim \operatorname{Col} A$ (the number of pivot columns).

Rank-nullity theorem

Theorem (Rank-nullity theorem). If A is an $m \times n$ matrix, then

$$\operatorname{rank} A + \dim \operatorname{Nul} A = n.$$

Proof: Row reduce A to $\operatorname{RREF}(A)$. The solutions to $A\vec{x} = \vec{0}$ are the same as the solutions to $\operatorname{RREF}(A)\vec{x} = \vec{0}$ (row equivalence preserves solutions). In $\operatorname{RREF}(A)$, each column is either:

- a *pivot column*: corresponds to a **basic variable**, determined by the free variables; or
- a *non-pivot column*: corresponds to a **free variable**, which can take any value in $\mathbb{R}$.

The number of pivot columns is $\operatorname{rank} A$ (since the pivot columns of A form a basis for $\operatorname{Col} A$, the number of pivots is $\dim \operatorname{Col} A$). The number of non-pivot (free) columns is $n - \operatorname{rank} A$.

Each free variable gives one basis vector for $\operatorname{Nul} A$ (obtained by setting that free variable to 1 and the other free variables to 0). Thus $\dim \operatorname{Nul} A$ = number of free variables = number of non-pivot columns = $n - \operatorname{rank} A$. Therefore

$$\operatorname{rank} A + \dim \operatorname{Nul} A = \operatorname{rank} A + (n - \operatorname{rank} A) = n.$$

■

Invertibility characterization via rank

Corollary. For an $n \times n$ matrix A, the following are equivalent:

1. A is invertible.
2. The columns of A form a basis for $\mathbb{R}^n$.
3. $\operatorname{rank} A = n$.
4. $\dim \operatorname{Nul} A = 0$.

Proof: (1) $\Leftrightarrow$ (2): A is invertible iff its columns are linearly independent and span $\mathbb{R}^n$ (by the full characterization theorem for square matrices), which is exactly the definition of the columns forming a basis for $\mathbb{R}^n$.

(2) $\Leftrightarrow$ (3): The columns form a basis for $\mathbb{R}^n$ iff they are linearly independent and span $\mathbb{R}^n$. By the proposition that pivot columns of A form a basis for $\operatorname{Col} A$, $\operatorname{rank} A = \dim \operatorname{Col} A$. So the columns span $\mathbb{R}^n$ iff $\operatorname{Col} A = \mathbb{R}^n$ iff $\dim \operatorname{Col} A = n$ iff $\operatorname{rank} A = n$. And if $\operatorname{rank} A = n$, all n columns are pivot columns, so the columns are linearly independent. Hence (2) $\Leftrightarrow$ (3).

(3) $\Leftrightarrow$ (4): By the rank-nullity theorem, $\operatorname{rank} A + \dim \operatorname{Nul} A = n$. So $\operatorname{rank} A = n$ iff $\dim \operatorname{Nul} A = 0$. ■

Proposition. If $U \subseteq V$ are subspaces of $\mathbb{R}^n$, then $\dim U \leq \dim V \leq n$. Moreover, if $U \subsetneq V$ then $\dim U < \dim V$, and if $\dim U = \dim V$ then $U = V$.

Proof: Let $\dim V = k$ with basis $\mathcal{B}_V = (\vec{b}_1, \ldots, \vec{b}_k)$. Since $U \subseteq V$, any linearly independent set in U is also a linearly independent set in $V \cong \mathbb{R}^k$ (via the coordinate map). By the proposition "$p > n \Rightarrow$ linearly dependent," any list of more than k vectors in V is dependent. Hence any basis of U has at most k elements: $\dim U \leq k = \dim V$. Also $\dim V \leq n$ since $V \subseteq \mathbb{R}^n$ and any basis of V is an independent list in $\mathbb{R}^n$ of size at most n.

If $U \subsetneq V$: pick $\vec{v} \in V \backslash U$. If $\mathcal{B}_U = (\vec{u}_1, \ldots, \vec{u}_l)$ is a basis of U, then $\vec{v} \notin \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_l\}$, so $\{\vec{u}_1, \ldots, \vec{u}_l, \vec{v}\}$ is linearly independent in V. Thus $l + 1 \leq \dim V$, i.e., $\dim U < \dim V$.

If $\dim U = \dim V = k$: let $\mathcal{B}_U = (\vec{u}_1, \ldots, \vec{u}_k)$ be a basis of U. This is a list of $k = \dim V$ linearly independent vectors in V. By the Basis theorem (part 1), this set is a basis for V. So $U = \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\} = V$. ■

6 Determinants

6.1 Defining Properties of the Determinant

Definition. The *determinant* is the unique function $\det : \{n \times n \text{ matrices}\} \to \mathbb{R}$ satisfying:

1. $\det I_n = 1$.
2. Interchanging two columns reverses the sign:
$$\det(\cdots \quad \vec{a}_i \quad \cdots \quad \vec{a}_j \quad \cdots) = -\det(\cdots \quad \vec{a}_j \quad \cdots \quad \vec{a}_i \quad \cdots).$$
3. Fixing all columns except the first, det is a linear function of the first column:
$$\det(\vec{u}+\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) = \det(\vec{u} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) + \det(\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n),$$
$$\det(c\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) = c \cdot \det(\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n).$$

Linearity in any column

Property (3) extends to any column via property (2): det is linear as a function of any single column.

6.2 Key Lemmas

Lemma. If A has two equal columns, then $\det A = 0$.

Proof: Suppose columns i and j of A are equal. Swapping these two columns produces the same matrix A (since they are equal). But by defining property (2), swapping two columns negates the determinant. Let A' be the matrix after the swap. Since $A' = A$, we have
$$\det A = \det A' = -\det A.$$
Adding $\det A$ to both sides gives $2\det A = 0$, so $\det A = 0$. ■

Corollary. If the columns of A are linearly dependent, then $\det A = 0$.

Proof: If the columns of A are linearly dependent, then one column, say column j, is a linear combination of the others: $\vec{a}_j = c_1\vec{a}_1 + \cdots + c_{j-1}\vec{a}_{j-1} + c_{j+1}\vec{a}_{j+1} + \cdots + c_n\vec{a}_n$. By linearity of det in column j (property (3)), $\det A$ equals det of the matrix with $\vec{a}_j$ replaced by each term. For the term $c_i\vec{a}_i$ $(i \neq j)$, the resulting matrix has two equal columns (columns i and j are both $c_i\vec{a}_i$). By the lemma, this contributes 0. Summing all such contributions, $\det A = 0$. ■

Corollary. If A is not invertible, then $\det A = 0$.

Proof: If A is not invertible, then (by the full characterization for square matrices) its columns are linearly dependent. By the previous corollary, $\det A = 0$. ■

6.3 Permutation Matrices

Definition. A *permutation matrix* is an $n \times n$ matrix formed by rearranging the columns of I_n, with entries in $\{0, 1\}$ having exactly one nonzero entry in each row and each column. Let S_n be the set of $n \times n$ permutation matrices. $|S_n| = n!$. If A is $n \times n$ and $X \in S_n$, then AX permutes the columns of A.

Definition. For $X \in S_n$, define $\text{inv}(X)$ as the number of 2×2 submatrices of X equal to $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Lemma. If $X \in S_n$, then $\det X = (-1)^{\text{inv}(X)}$.

Proof: We prove this by induction on $\text{inv}(X)$. The base case is $\text{inv}(X) = 0$, which means X has no $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ submatrix; this forces $X = I_n$ (the only permutation matrix with no inversions is the identity). By property (1), $\det I_n = 1 = (-1)^0$.

For the inductive step: suppose $\text{inv}(X) = k > 0$. Then $X \neq I_n$, so X has a column j that is not $\vec{e}_j$. There must exist a 2×2 submatrix of the form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; concretely, there are columns $i < j$ of X that are "out of order." Let X' be obtained from X by swapping two adjacent columns that are out of order (i.e., swap columns $i, i+1$ where column i has its 1 in a row below the 1 of column $i+1$). This swap changes inv by exactly ± 1 (it changes the status of exactly one 2×2 submatrix), so $\text{inv}(X') = k - 1$. By the induction hypothesis, $\det X' = (-1)^{k-1}$. By property (2), swapping two columns negates the determinant, so $\det X = -\det X' = -(-1)^{k-1} = (-1)^k = (-1)^{\text{inv}(X)}$. ■

Lemma. If X has 0/1 entries with one nonzero per column but is **not** a permutation matrix (i.e., $X \notin S_n$), then $\det X = 0$.

Proof: If X has 0/1 entries with one nonzero entry per column but $X \notin S_n$, then by pigeonhole some row must contain two or more nonzero entries (since n columns with one nonzero each, but not one per row, forces a collision). So there exist columns i, j and a row r where $X_{ri} = X_{rj} = 1$. Since each column has exactly one nonzero entry, this means two columns of X have their 1 in the same row. But two columns being $\vec{e}_r$ means they are equal — contradicting the condition that X is a permutation matrix. Hence the columns of X are linearly dependent (two equal columns), so by the lemma (two equal columns $\Rightarrow \det = 0$), $\det X = 0$.

More precisely: the columns of X are a list of n standard basis vectors (with one nonzero entry each). If two columns have their 1 in the same row, those two columns are equal, so X has two equal columns, giving $\det X = 0$ by the lemma. ■

6.4 General Formula for $\det A$

Definition. For a matrix X with 0/1 entries and a matrix A, define $\text{prod}(X, A)$ as the product of the entries of A in the nonzero positions of X.

Determinant formula

Theorem. For any $n \times n$ matrix A,

$$\det A = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}.$$

Proof: Let $A = \begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}$ with each $\vec{a}_j = \sum_{i=1}^{n} a_{ij}\vec{e}_i$. By multilinearity (property (3) applied to each column), $\det A$ expands as

$$\det A = \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \cdot \det \begin{pmatrix} \vec{e}_{i_1} & \vec{e}_{i_2} & \cdots & \vec{e}_{i_n} \end{pmatrix}.$$

The matrix $\begin{pmatrix} \vec{e}_{i_1} & \cdots & \vec{e}_{i_n} \end{pmatrix}$ is a 0/1 matrix with one nonzero per column. If it is a permutation matrix $X \in S_n$ (i.e., the $i_1, \ldots, i_n$ are all distinct), then by the lemma $\det X = (-1)^{\text{inv}(X)}$; otherwise (two indices equal) it has two equal rows and $\det = 0$ by

the lemma on non-permutation 0/1 matrices. So only the $X \in S_n$ terms survive, and for those, $a_{i_1 1} \cdots a_{i_n n} = \text{prod}(X, A)$. Thus

$$\det A = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}.$$

■

Theorem. The determinant is the unique function with properties (1) $\det I_n = 1$, (2) column swaps negate the determinant, (3) linearity in each column.

Proof: *Existence:* The formula $\det A = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}$ (previous theorem) defines a function. One verifies it satisfies properties (1)–(3): (1) For $A = I_n$, $\text{prod}(X, I_n) = 0$ for $X \neq I_n$ (some entry of I_n outside X's nonzero positions is needed), and $\text{prod}(I_n, I_n) = 1$ with $\text{inv}(I_n) = 0$, so $\det I_n = 1$. (2) Swapping columns i, j of A swaps the corresponding rows of each X, which changes $\text{inv}(X)$ by an odd number (sign flip), giving the $-$ sign. (3) $\text{prod}(X, A)$ is linear in each column of A, hence so is the sum.

Uniqueness: Suppose f and g both satisfy (1)–(3). By the same multilinear expansion as in the previous proof, any such f must satisfy

$$f(A) = \sum_{X \in S_n} \text{prod}(X, A) \cdot f(X).$$

By the lemma, $f(X) = (-1)^{\text{inv}(X)}$ for $X \in S_n$ (the proof of that lemma uses only properties (1) and (2), which f satisfies). So $f(A) = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}$, which is the same for any f. Hence $f = g = \det$. ■

Small cases.

- $n = 1$: $\det(a) = a$.
- $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
- $n = 3$: $\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$.

Geometric meaning of determinant

For a 2×2 matrix, $|\det A|$ is the area of the parallelogram spanned by the columns of A. For $n \times n$, $|\det A|$ is the volume of the n-dimensional parallelepiped spanned by the columns.

6.5 Determinant of Transpose

Lemma. If $X \in S_n$ then $X^T \in S_n$ and $\text{inv}(X) = \text{inv}(X^T)$.

Proof: If $X \in S_n$, then X has exactly one nonzero entry in each row and each column. Transposing swaps rows and columns, so X^T also has exactly one nonzero entry in each row and each column. Hence $X^T \in S_n$.

For the inversion count: a 2×2 submatrix of X at rows $i < j$ and columns $k < l$ equals $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ iff in X, rows i, j and columns k, l have entries $X_{il} = X_{jk} = 1$ and $X_{ik} = X_{jl} = 0$. Under transpose, this submatrix becomes the submatrix of X^T at rows $k < l$

and columns $i < j$, with entries $(X^T)_{ki} = X_{ik} = 0$, $(X^T)_{lj} = X_{jl} = 0$, $(X^T)_{li} = X_{il} = 1$, $(X^T)_{kj} = X_{jk} = 1$. So the transposed submatrix is also $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Thus transposition gives a bijection between the inversion submatrices of X and of X^T, so $\text{inv}(X) = \text{inv}(X^T)$. ■

Determinant of transpose

Corollary. $\det A = \det(A^T)$ for any square matrix A.

Proof: Using the general formula,

$$\det(A^T) = \sum_{X \in S_n} \text{prod}(X, A^T) \cdot (-1)^{\text{inv}(X)}.$$

Since $X \in S_n \Leftrightarrow X^T \in S_n$ and $\text{inv}(X) = \text{inv}(X^T)$ (previous lemma), we can re-index the sum by $Y = X^T$:

$$\det(A^T) = \sum_{Y \in S_n} \text{prod}(Y^T, A^T) \cdot (-1)^{\text{inv}(Y)}.$$

Now $\text{prod}(Y^T, A^T)$ is the product of entries of A^T in the nonzero positions of Y^T. But the nonzero positions of Y^T are the transposes of the nonzero positions of Y, and $(A^T)_{ji} = A_{ij}$. So $\text{prod}(Y^T, A^T) = \text{prod}(Y, A)$. Hence

$$\det(A^T) = \sum_{Y \in S_n} \text{prod}(Y, A) \cdot (-1)^{\text{inv}(Y)} = \det A.$$

■

Corollary. If A has two equal rows, then $\det A = 0$.

Proof: If A has two equal rows, then A^T has two equal columns. By the lemma (two equal columns $\Rightarrow$ det $= 0$), $\det(A^T) = 0$. By the corollary $\det A = \det(A^T)$, so $\det A = 0$. ■

6.6 Determinant of Products

$\det(AB) = (\det A)(\det B)$

Lemma (case $\det A \neq 0$). Define $f(M) = \dfrac{\det(AM)}{\det A}$. Then f satisfies the three defining properties of the determinant, so $f = \det$. Thus $\dfrac{\det(AB)}{\det A} = \det B$.

Theorem. For any $n \times n$ matrices A and B: $\det(AB) = (\det A)(\det B)$.

Lemma. If A or B is not invertible, then AB is not invertible.

Proof of the lemma (case $\det A \neq 0$): We verify $f(M) = \dfrac{\det(AM)}{\det A}$ satisfies the three defining properties of the determinant (as a function of M):

1. $f(I_n) = \dfrac{\det(AI_n)}{\det A} = \dfrac{\det A}{\det A} = 1$.
2. Swapping two columns i, j of M swaps the corresponding columns of AM (since $AM = (A\vec{m}_1 \ \cdots \ A\vec{m}_n)$ depends column-wise on M). By property (2) of det, $\det(AM') = -\det(AM)$, so $f(M') = -f(M)$.

3. Fixing all columns of M except the first: $AM = (A\vec{m}_1 \quad A\vec{m}_2 \quad \cdots \quad A\vec{m}_n)$. If $\vec{m}_1 = \vec{u} + \vec{v}$, then $A\vec{m}_1 = A\vec{u} + A\vec{v}$, and by linearity of det in column 1, $\det(AM_{\vec{u}+\vec{v}}) = \det(AM_{\vec{u}}) + \det(AM_{\vec{v}})$. Dividing by $\det A$, f is linear in column 1. Similarly for scalar multiplication.

Since f satisfies (1)–(3), by the uniqueness theorem $f = \det$. Hence $\dfrac{\det(AB)}{\det A} = f(B) = \det B$, giving $\det(AB) = (\det A)(\det B)$ when $\det A \neq 0$. ■

Proof of the lemma (non-invertible $\Rightarrow$ AB non-invertible): If A is not invertible, then the columns of A do not span $\mathbb{R}^n$ (by the full characterization for square matrices), so $\operatorname{Col} A \subsetneq \mathbb{R}^n$. Then $\operatorname{Col}(AB) = \{AB\vec{x} : \vec{x} \in \mathbb{R}^n\} = \{A\vec{y} : \vec{y} \in \operatorname{Col} B\} \subseteq \operatorname{Col} A \subsetneq \mathbb{R}^n$. So the columns of AB do not span $\mathbb{R}^n$, hence AB is not invertible.

If B is not invertible, then the columns of B are linearly dependent. So there exists $\vec{x} \neq \vec{0}$ with $B\vec{x} = \vec{0}$. Then $(AB)\vec{x} = A(B\vec{x}) = A\vec{0} = \vec{0}$, so the columns of AB are also linearly dependent, hence AB is not invertible. ■

Proof of the theorem: *Case 1:* $\det A \neq 0$. By the lemma (case $\det A \neq 0$), $\det(AB) = (\det A)(\det B)$.

Case 2: $\det A = 0$. Then A is not invertible. By the lemma, AB is not invertible, so by the corollary (non-invertible $\Rightarrow$ $\det = 0$), $\det(AB) = 0$. And $(\det A)(\det B) = 0 \cdot (\det B) = 0$. So $\det(AB) = 0 = (\det A)(\det B)$.

In both cases, $\det(AB) = (\det A)(\det B)$. ■

6.7 Triangular Matrices

Definition. A matrix is *upper-triangular* if all entries below the main diagonal are zero. It is *lower-triangular* if all entries above the main diagonal are zero. A matrix is *triangular* if it is either upper- or lower-triangular. A matrix is *diagonal* if it is both upper- and lower-triangular (nonzero entries only on the diagonal).

Determinant of triangular matrices

Proposition. If A is triangular, then $\det A$ is the product of the diagonal entries: $\det A = a_{11}a_{22}\cdots a_{nn}$.

Proof: *Case 1: A is upper-triangular* (so $a_{ij} = 0$ for $i > j$). Using the general formula $\det A = \sum_{X \in S_n} \operatorname{prod}(X, A) \cdot (-1)^{\operatorname{inv}(X)}$, we analyze each term. For $X = I_n$, $\operatorname{prod}(I_n, A) = a_{11}a_{22}\cdots a_{nn}$ (the diagonal entries) and $(-1)^{\operatorname{inv}(I_n)} = 1$.

For $X \neq I_n$ (any other permutation matrix), X has a nonzero entry in some position (i, j) with $i > j$ (i.e., below the diagonal) — because X being different from I_n means some column's 1 is not on the diagonal, and by counting (since each row and column has exactly one 1), at least one such 1 must be below the diagonal. Then $\operatorname{prod}(X, A)$ includes the factor a_{ij} with $i > j$, which is 0 (since A is upper-triangular). So $\operatorname{prod}(X, A) = 0$ for $X \neq I_n$.

Therefore $\det A = \operatorname{prod}(I_n, A) \cdot 1 = a_{11}a_{22}\cdots a_{nn}$.

Case 2: A is lower-triangular. Then A^T is upper-triangular, and by Case 1, $\det(A^T) = a_{11}a_{22}\cdots a_{nn}$ (the diagonal of A^T equals the diagonal of A). By $\det A = \det(A^T)$, we get $\det A = a_{11}a_{22}\cdots a_{nn}$. ■

6.8 Invertibility and Determinant

A invertible $\Leftrightarrow \det A \neq 0$

Lemma. $\det A$ is a nonzero multiple of $\det(\mathrm{RREF}(A))$. Each row operation multiplies det by a nonzero scalar:
- Row swap: multiply by -1.
- Row scaling by $\lambda \neq 0$: multiply by λ.
- Row replacement (add multiple of one row to another): no change.

Theorem. A is invertible if and only if $\det A \neq 0$.

Proof of the lemma: By $\det A = \det(A^T)$, the effect of a **row** operation on det equals the effect of the corresponding **column** operation. We know the column operations behave as:

- *Column swap*: by defining property (2), this multiplies det by -1. Hence row swap multiplies det by -1.
- *Column scaling by* $\lambda \neq 0$: by linearity (property (3)),
$$\det(\cdots \quad \lambda\vec{a}_j \quad \cdots) = \lambda \det(\cdots \quad \vec{a}_j \quad \cdots).$$
Hence row scaling multiplies det by $\lambda \neq 0$.
- *Column replacement* (add λ times column j to column i): by linearity in column i,
$$\det(\cdots \quad \vec{a}_i + \lambda\vec{a}_j \quad \cdots) = \det(\cdots \quad \vec{a}_i \quad \cdots) + \lambda \det(\cdots \quad \vec{a}_j \quad \cdots).$$
The second term has two equal columns (columns i and j both equal $\vec{a}_j$), so it is 0. Hence det is unchanged. The same holds for row replacement.

Since each row operation multiplies det by a nonzero scalar (-1 or $\lambda \neq 0$ or 1), the overall factor between $\det A$ and $\det(\mathrm{RREF}(A))$ is a product of nonzero scalars, hence nonzero. ■

Proof of the theorem: ($\Leftarrow$) Suppose $\det A \neq 0$. Write $R = \mathrm{RREF}(A)$. By the lemma, $\det A = c \cdot \det R$ for some nonzero scalar c. So $\det R = (\det A)/c \neq 0$. Now R is in reduced row echelon form. If R had a row of zeros (no pivot in that row), then by the proposition on triangular matrices (since R is upper-triangular with some zero on the diagonal), $\det R = 0$, a contradiction. So every row has a pivot. Since R is $n \times n$ square and has n pivots (one per row, and at most one per column, so exactly n), we have $R = I_n$. Hence A is row equivalent to I_n, so A is invertible.

($\Rightarrow$) Suppose A is invertible. Then $\mathrm{RREF}(A) = I_n$, so $\det(\mathrm{RREF}(A)) = \det I_n = 1 \neq 0$. By the lemma, $\det A = c \cdot \det(\mathrm{RREF}(A)) = c \cdot 1 = c$ for some nonzero c, so $\det A \neq 0$.

(Equivalently, for ($\Rightarrow$): if A is invertible, its columns are linearly independent. By the corollary (dependent columns $\Rightarrow$ det $= 0$), the contrapositive gives $\det A \neq 0$.) ■

Corollary. If A has a zero row or a zero column, then $\det A = 0$.

Proof: If A has a zero column, then the columns of A are linearly dependent (any set containing the zero vector is linearly dependent). By the corollary (linearly dependent columns $\Rightarrow$ det $= 0$), $\det A = 0$.

If A has a zero row, then A^T has a zero column. By the above argument, $\det(A^T) = 0$. Since $\det A = \det(A^T)$, we get $\det A = 0$. ■

Corollary. For any $n \times n$ matrix A and scalar $c \in \mathbb{R}$: $\det(cA) = c^n \det A$.

Proof: The matrix cA is obtained from A by scaling each of the n rows by c. By the lemma on row operations, scaling a single row by c multiplies the determinant by c. Applying this n times (once for each row), we get

$$\det(cA) = \underbrace{c \cdot c \cdots c}_{n \text{ times}} \det A = c^n \det A.$$

(Equivalently: cA scales each of the n columns by c, and by multilinearity (property (3)) in each column, each scaling contributes a factor of c, giving $c^n \det A$.) ■

Corollary. If A is invertible, then $\det(A^{-1}) = \dfrac{1}{\det A}$.

Proof: Since A is invertible, A^{-1} exists and $AA^{-1} = I_n$. By the theorem $\det(AB) = (\det A)(\det B)$, we have

$$\det(AA^{-1}) = (\det A)(\det A^{-1}).$$

But $AA^{-1} = I_n$, so $\det(AA^{-1}) = \det I_n = 1$. Therefore

$$(\det A)(\det A^{-1}) = 1.$$

Since A is invertible, $\det A \neq 0$ (by the previous theorem). Dividing both sides by $\det A$ gives

$$\det(A^{-1}) = \frac{1}{\det A}.$$

■

6.9 Computing Determinants

Algorithm 1: Row Reduction

Row reduction algorithm for $\det A$

1. Set `denom` $= 1$.
2. Row reduce A to echelon form E (upper-triangular suffices). Update `denom`:
   - Row swap: `denom` $\leftarrow -$`denom`.
   - Scale row by $\lambda \neq 0$: `denom` $\leftarrow \lambda \cdot$ `denom`.
   - Row replacement: no change to `denom`.
3. $\det E =$ product of diagonal entries of E.
4. $\det A = \dfrac{\det E}{\texttt{denom}}$.

Algorithm 2: Cofactor Expansion

Definition. Let $A_{(i,j)}$ be the submatrix formed by deleting row i and column j from A.

Cofactor expansion

Theorem. For an $n \times n$ matrix A with entry a_{ij} in row i, column j:

$$\det A = a_{11} \det A_{(1,1)} - a_{12} \det A_{(1,2)} + a_{13} \det A_{(1,3)} - \cdots + (-1)^{1+n} a_{1n} \det A_{(1,n)}$$

$$\det A = a_{11} \det A_{(1,1)} - a_{21} \det A_{(2,1)} + a_{31} \det A_{(3,1)} - \cdots + (-1)^{n+1} a_{n1} \det A_{(n,1)}$$

(Expansion along the first row and first column, respectively.)

Proof: We prove the row expansion (expansion along the first row); the column expansion follows from $\det A = \det(A^T)$.

Write A with columns $\vec{a}_1, \ldots, \vec{a}_n$. By linearity of det in the first column, write $\vec{a}_1 = a_{11}\vec{e}_1 + a_{21}\vec{e}_2 + \cdots + a_{n1}\vec{e}_n$. Then

$$\det A = \sum_{i=1}^{n} a_{i1} \det \begin{pmatrix} \vec{e}_i & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}.$$

Let $M_i = \begin{pmatrix} \vec{e}_i & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}$. We compute $\det M_i$.

In M_i, the first column is $\vec{e}_i$. If $i \neq 1$, swap row 1 with row i (using $i-1$ adjacent row swaps) to move the 1 to the top-left corner. Each swap multiplies det by -1, so $\det M_i = (-1)^{i-1} \det M_i'$ where M_i' has first column $\vec{e}_1$ and the other columns are $\vec{a}_2, \ldots, \vec{a}_n$ (with rows 1 and i swapped). Now in M_i', the first row is $\begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}$. By the general formula, only the term $X = I_n$ (or X with first column $= \vec{e}_1$) contributes, and the remaining determinant is exactly that of the submatrix $A_{(i,1)}$ (delete row i and column 1 from A, with appropriate relabeling). So $\det M_i' = \det A_{(i,1)}$, giving $\det M_i = (-1)^{i-1} \det A_{(i,1)}$.

Hence $\det A = \sum_{i=1}^{n} a_{i1}(-1)^{i-1} \det A_{(i,1)}$, which is the expansion along column 1:

$$\det A = a_{11} \det A_{(1,1)} - a_{21} \det A_{(2,1)} + a_{31} \det A_{(3,1)} - \cdots + (-1)^{n+1} a_{n1} \det A_{(n,1)}.$$

For expansion along the first row: apply the column expansion to A^T (whose first column is the first row of A), then use $\det A = \det(A^T)$:

$$\det A = \det(A^T) = \sum_{j=1}^{n} a_{1j}(-1)^{j-1} \det(A^T)_{(j,1)} = \sum_{j=1}^{n} (-1)^{j+1} a_{1j} \det A_{(1,j)},$$

since $(A^T)_{(j,1)} = (A_{(1,j)})^T$ and $\det(A_{(1,j)})^T = \det A_{(1,j)}$. ■

When to use which algorithm

- Row reduction: efficient for general matrices, especially $n > 3$.
- Cofactor expansion: useful when A has many zero entries (few terms) or $n \leq 3$.

Example. For $n = 3$:

$$\begin{aligned} \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} &= a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg). \end{aligned}$$

Summary of determinant properties

- $\det I_n = 1$.
- $\det A = \det(A^T)$.
- $\det(AB) = (\det A)(\det B)$.
- A invertible $\Leftrightarrow \det A \neq 0$.
- If A is triangular, $\det A =$ product of diagonal entries.
- $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$; $(A^T)^{-1} = (A^{-1})^T$.